



TECHNICAL SPECIFICATIONS ON NAPAS SYSTEM CONNECTIVITY <Apply for Member Institution>

PART I: TRANSACTION PROCESING

NATIONAL PAYMENT CORPORATION OF VIETNAM

Ha Noi, December, 2016



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REVISION HISTORY

Version	Release Date	Editor	Reviewer	Description for change
0.9	From 01/2016 - 11/2016	Ha Nam Ninh Nguyen Hung Cuong Bui Thi Kim Dung Pham Minh Ngoc Le Anh Tuan Dao Thanh Son Huynh Cong Linh Nguyen Thanh Quynh	Nguyen Hung Nguyen	- Built a draft Technical Specification Document - Spitted the document to 05 parts include: • Part I – Transaction Processing • Part II – Message Format • Part III – Reconciliation File Format • Part IV – Security and Communication • Part V – Annex - Added international transaction processing on following part: • Part VI – Annex: International Transaction Processing
1.0	12/2016	Ha Nam Ninh Nguyen Hung Cuong Nguyen Thanh Quynh Dao Thanh Son Huynh Cong Linh	Nguyen Hung Nguyen	- Added some data elements on CHIP transaction at: • Part II – Message Format: Section 6 – Data elements, Section 7 – Message format • Part V – Annex: Section 4 –

				Response code - Updated the Online switching (Ecom), Tokenization process flow synchronized with regulations for member organizations participating in the NAPAS system at: • Part I – Transaction processing, Section 4.2, 4.3, 4.5 • Part II – Message Format, Section 6.2 - Added service and interchange fees between NAPAS and Member Banks; added two fields : DE 22, DE 25 at : • Part III – Message structure : section 4.3
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1 Audience

“Technical Specifications on NAPAS System Connectivity” document is intended for the Member Organizations (MO) of NAPAS include but not limited to Member Banks of NAPAS and Payment Intermediaries that directly connected to card switching services.

2 Scope

The card switching service allows customers to use Member Banks card in NAPAS network to perform transactions on the device / card accepted channels of NAPAS Member Institution as well as International Card Association or Switching Organizations of other countries and vice versa.

This document describes the transaction processing of the switching services that NAPAS deploys to Member Institutions.

Switching services include but not limited to:

2.1 Domestic card switching service:

- a. Domestic card switching services at ATM: allow customers to use Member Banks card to perform transactions on the ATM equipment of the Member Banks in the Switching Organizations of other countries, and vice versa. Domestic card switching services on the ATM include but not limited to the following transactions:
 - Balance Inquiry
 - Cash Withdrawal
 - InTra-bank Fund Transfer (ITFT)
 - Mini Statement
 - PIN Change
- b. Domestic card switching service at POS: allows customers to use Member Banks card to perform transactions at POS of Member Bank of NAPAS network or Member Bank of Switching Organizations of other countries, and vice versa. Card switching services at POS include but not limited to the following transactions:
 - Purchase
 - Void
 - Balance Inquiry

- PIN Change
- c. Online switching service (Ecom): allows customers to make purchase through the online payment website that is connected to NAPAS. A successful payment transaction consists of 03 messages after sending to Member Bank:
 - Verify Cardholder information (Verify Card)
 - Verify OTP
 - Ecom Purchase

Beside normally online payment service, NAPAS also debits customer account by processing only one message:

- Ecom Fast Purchase
- d. Inter-Bank Fund Transfer 24/7 service (IBFT): allows the customers to transfer money on many different payment channels (ATM, Internet banking, Mobile banking, Bank counters ...). In addition, this is also a service allows Payment Intermediaries to make a fund transfer from e-wallet account of Payment Intermediaries to account of Member Bank of NAPAS. This service includes following messages:
 - Beneficiary cardholder / account information inquiry (IBFT Inquiry)
 - Beneficiary cardholder / account fund transfer (IBFT Deposit)

Note: IBFT Inquiry message and IBFT Deposit message are 02 independent messages and NAPAS system does not check the relationship between these messages.

- e. Payment code service: allows customer to use Member Bank payment channels which transfers and receives using Payment code (PC) on the transaction. Customer who transfers money is provided a Payment code when performs transfer transaction, beneficiary customer uses this Payment code on channel of Bank to receive money. On-us transfer or off-us transfer bank and receive bank may be on the same bank or two different banks. The beneficiary customer object of service mainly directs to customers who have not got bank account.
 - Payment code create request (PC Create)
 - Payment code authentication (PC Verify)
 - Payment code withdrawal (PC Cash Withdrawal)
 - Payment code inquiry (PC Inquiry)

- f. Tokenization service: the service that encrypts customer information to Token ID range – a range codes to identify alternative using of customer information when conducting payment.
- Card information verify (Token Verify Card)
 - OTP verify (Token Verify OTP)
 - Token Purchase
 - Token Fast Purchase

2.2 Cross-border card switching services:

- a. Cross-border card switching services at ATM/POS: allows customer to use the card of NAPAS Member Bank to perform transactions on the Member Banks ATM/POS equipment of the Switching Organization of other countries, and vice versa. Cross-border card switching services at ATM/POS include but not limited to the following transactions:
- Balance Inquiry at ATM
 - Cash Withdrawal at ATM
 - Balance Inquiry at POS
 - Purchase at POS
- b. Cross-Border Fund Transfer – CBFT: allows customers to use their bank cards in Switching Organizations of other countries can transfer money of account number/client card number of Member bank in NAPAS. CBFT service deploys to Member Bank based on the platform of the IBFT service, includes two transactions:
- Beneficiary cardholder / account information inquiry (CBFT Inquiry)
 - Beneficiary cardholder / account fund transfer (CBFT Deposit)

3 Terms and Acronyms

The following table describes the terminologies and acronyms used on this document:

No.	Terms/Acronyms	Description
1	SW	Switching system (Switch) of NAPAS
2	ACQ	Acquirer (Settlement Bank)
3	ISS	Issuer
4	BNB	Beneficiary Bank
5	PMI	Payment Intermediary
6	MO	Member organization (Include Member Bank and Payments Intermediary)
7	Bank MO	Bank Member Organization
8	ICA	International Card Association (include Visa, MasterCard, Amex, JCB, etc...)
9	ATM	Automated Teller Machine
10	POS	Point Of Sale/Service
11	ITFT	InTra-bank Fund Transfer (Transfer to internal bank)
12	IBFT	Inter-Bank Fund Transfer 24/7
13	CBFT	Cross-Border Fund Transfer
14	OTP	One Time Password
15	APN	Asian Payment Network
16	MEPS	Malaysia Electronic Payment System
17	KTFC	Korea Financial Telecommunications & Clearings Institute
18	NETS	Network For Electronic Transfer
19	ITMX	Interbank Transaction Management and Exchange
20	CUP	China Union Pay

21	NCC UC	Union Card
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4 Domestic card switching service transaction flow

4.1 Card switching service at ATM/ POS

4.1.1 Successful transaction flow

4.1.1.1 Card switching service at ATM

Card switching service at ATM includes the following transactions:

- Cash withdrawal
- Balance Inquiry
- Mini-statement
- Intra-Bank Fund Transfer
- Change pin (Pin Change)

Above transaction flows are described as detail following:

4.1.1.1.1 Cash Withdrawal

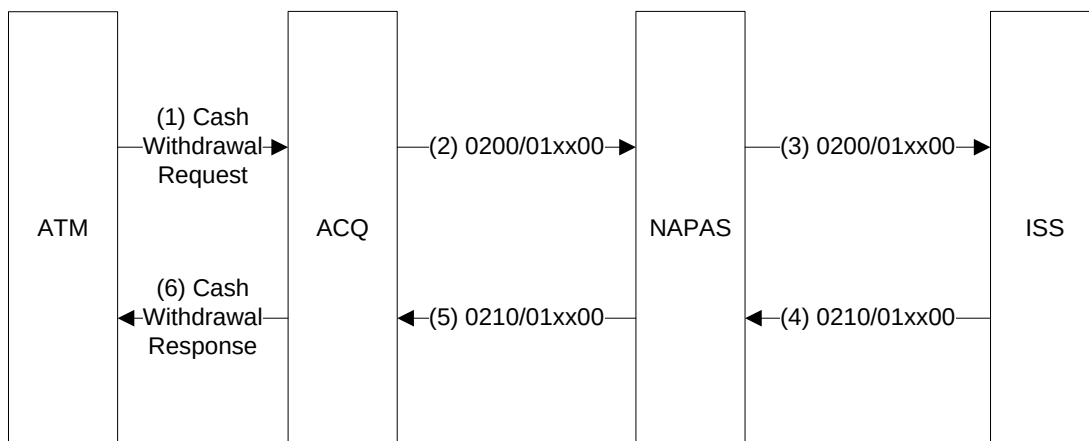


Figure 1. Cash withdrawal transaction flow at an ATM

Transaction flow detail:

Step	Transaction flow
1	Card holder inserts a card into an ATM and enters PIN number on the ATM screen. Selects "Cash withdrawal" button on the screen.

	ATM initiates a cash withdrawal request message to ACQ.
2	ACQ sends the cash withdrawal request message to NAPAS (Pcode 01xx00).
3	NAPAS forwards the message to ISS.
4	ISS processes and responds the message to NAPAS.
5	NAPAS forwards the response message to ACQ.
6	ACQ sends the response message to ATM.

4.1.1.1.2 Balance inquiry

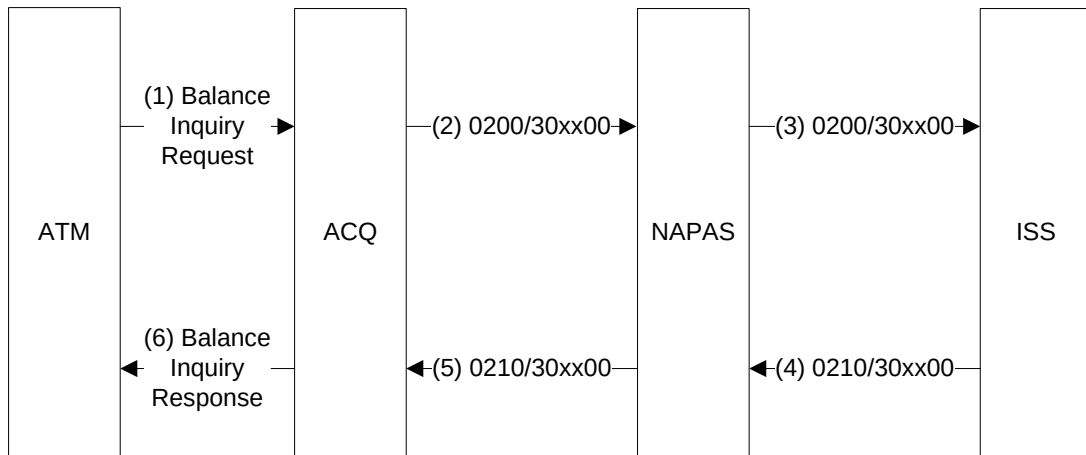


Figure 2. Balance inquiry transaction flow at an ATM

Transaction flow detail:

Step	Transaction flow
1	Card holder inserts a card into an ATM and enters PIN number on the ATM screen. Selects " Balance inquiry " function on the screen. ATM initiates a balance inquiry request message to ACQ.
2	ACQ sends the balance inquiry request message to NAPAS (Pcode 30xx00).
3	NAPAS forwards the request message to ISS.

4	ISS processes and responds the message to NAPAS.
5	NAPAS forwards the response message to ACQ.
6	ACQ sends the response message to ATM.

4.1.1.1.3 Mini-statement

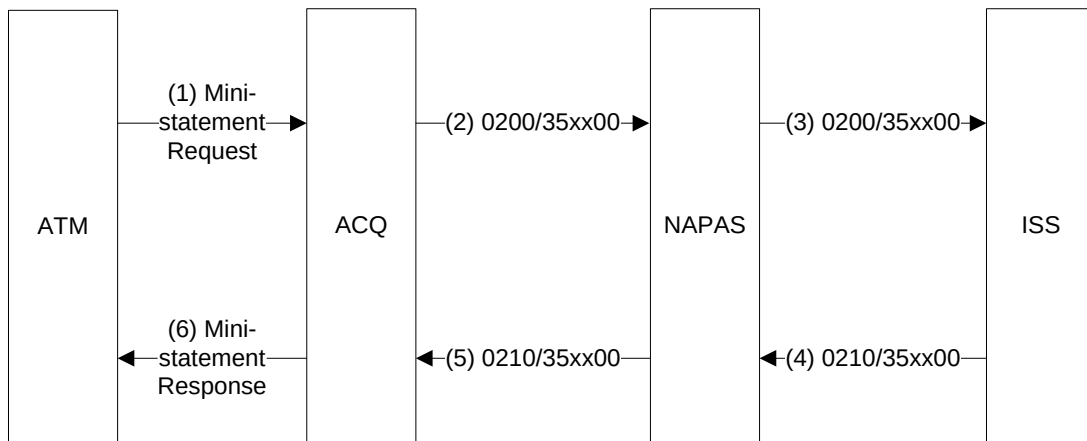


Figure 3. Mini-statement transaction flow at an ATM

Transaction flow detail:

Step	Transaction flow
1	Cardholder inserts a card into an ATM and enters PIN number on the screen ATM. Selects “ Mini-statement ” function on the screen. ATM initiates a mini-statement request message to ACQ
2	ACQ sends a mini-statement the request message to NAPAS (Pcode 35xx00).
3	NAPAS forwards the message to ISS
4	ISS processes and responds a message to NAPAS.
5	NAPAS forwards the response message to ACQ
6	ACQ sends the response message to ATM.

Note: for mini-statement transaction, the ATM will print up to 10 most recent transactions.

4.1.1.1.4 InTra-bank Fund transfer

InTra-bank fund transfer (ITFT) transaction allows customer of member bank to transfer from card to account number / other card of the same bank at ATM of other member banks.

Intra-bank fund transfer service includes only one transaction:

- InTra-bank fund transfer (ITFT Deposit)

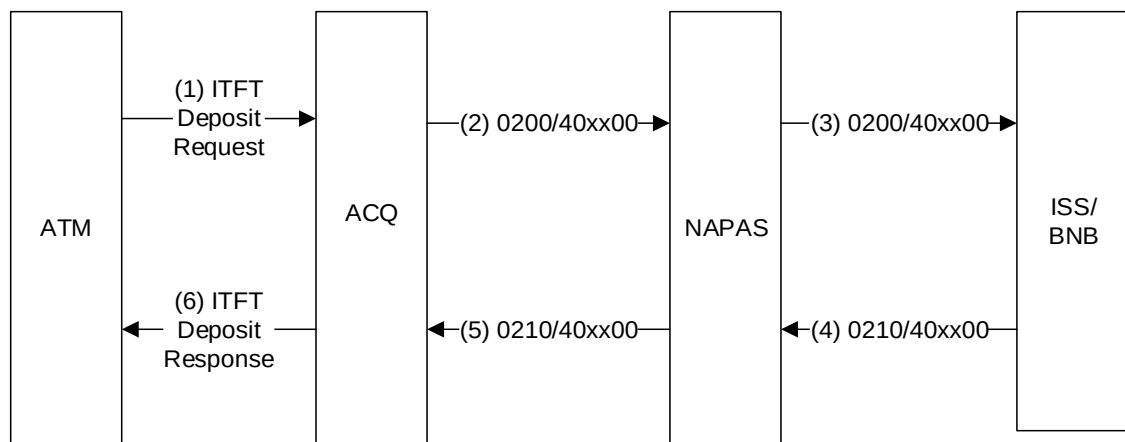


Figure 4. InTra-bank fund transfer transaction flow (ITFT Deposit)

Transaction flow detail:

Step	Transaction flow
1	Cardholder inserts a card into ATM and enters PIN number. Selects “ITFT Transaction” function on the ATM screen. Selects account type (Default, Saving, Current). Cardholder selects Beneficiary Bank Enters beneficiary account / beneficiary card number Enters transfer amount Cardholder confirms the transfer information on the screen.
2	ACQ sends an ITFT request message to NAPAS (Pcode 40xx00).

3	NAPAS forwards the ITFT request message to ISS/BNB
4	ISS/BNB processes and sends an ITFT response message to NAPAS.
5	NAPAS forwards the ITFT response message to ACQ.
6	ACQ sends the response message to ATM.

4.1.1.1.5 PIN Change

PIN change transaction allows cardholder of member bank to change personal identification number (PIN number) through ATM of other member banks.

PIN change request message (0200) is initiated from Acquiring Bank (ACQ) and sends it to NAPAS. ACQ routes this transaction based on NAPAS card range and forwards it to Issuing Bank.

ISS checks and resends a response message (0210) to ACQ through NAPAS.

In some cases, PIN change transaction was not successful at ACQ but the process of changing PIN at ISS was still successful. Therefore, ACQ should notify customer to re-check after performing PIN change.

PIN change transaction does not have a reversal transaction, if ACQ sends a reversal transaction to NAPAS, NAPAS will reject by a response message having RC <> 00.

- New PIN block value is stored in DE105 field (New PIN block needs to be encrypted by ZPK is same as origin PIN Block in DE52).
- Old PIN block is stored in DE52.

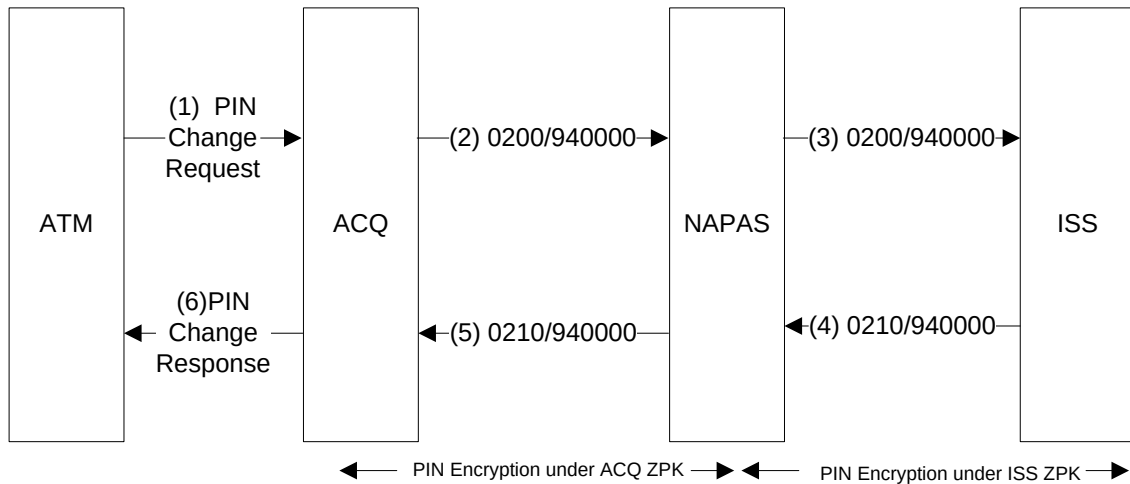


Figure 5. PIN change transaction flow

Transaction flow detail:

Step	Transaction flow
1	Selects “PIN change” function on the ATM screen. Cardholder performs to enter old PIN and select Enter button Cardholder performs to enter new PIN and select Enter button Cardholder performs to enter new PIN and select Enter button to confirm If customer enters PIN and confirm PIN are not the same then the ATM prompts again to re-enter new PIN. ATM initiates a PIN change request message to ACQ.
2	ACQ sends the PIN change request message to NAPAS (Pcode 940000).
3	NAPAS processes and forwards the message to ISS.
4	ISS checks old PIN of customer before perform changing new PIN and responds a message to NAPAS.
5	NAPAS forwards the response message to ACQ.
6	ACQ sends the response message to ATM.

Note: PIN change does not have a reversal transaction, in time-out case, ATM will notify customer to re-check the PIN could change or not:

“PIN change transaction may be processed unsuccessful, please re-check your PIN”

4.1.1.2 Card switching service at POS

Card switching service at POS includes transaction below:

- Balance inquiry
- Purchase
- Void
- PIN change

Above transaction flows are describes as detail following:

4.1.1.2.1 Balance Inquiry

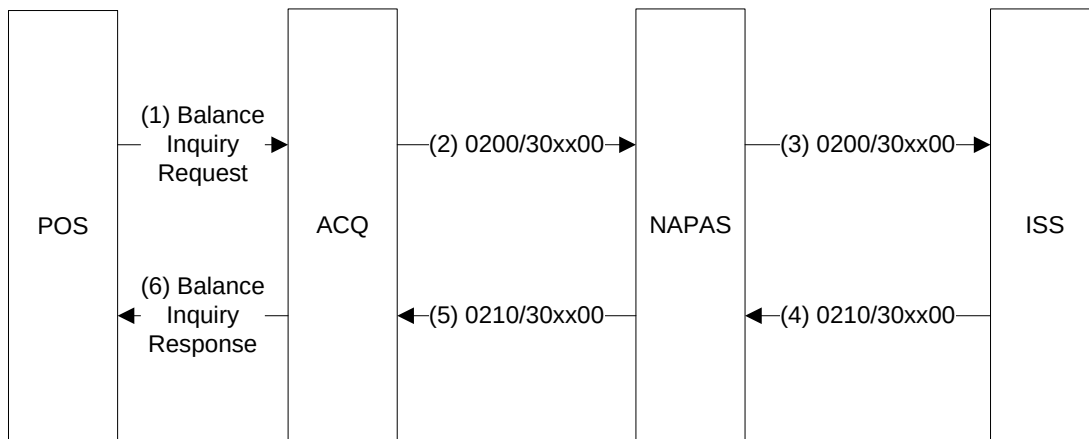


Figure 6. POS balance inquiry transaction flow

Transaction flow detail:

Step	Transaction flow
1	Cardholder selects “ Balance Inquiry ” function, swipes card and enters PIN number at POS POS initiates a balance inquiry request message to ACQ
2	ACQ sends the balance inquiry request message to NAPAS (Pcode 30xx00)
3	NAPAS forwards the message to ISS.

4	ISS processes and responds message to NAPAS.
5	NAPAS forwards the response message to ACQ.
6	ACQ sends the response message to POS.

4.1.1.2.2 Purchase

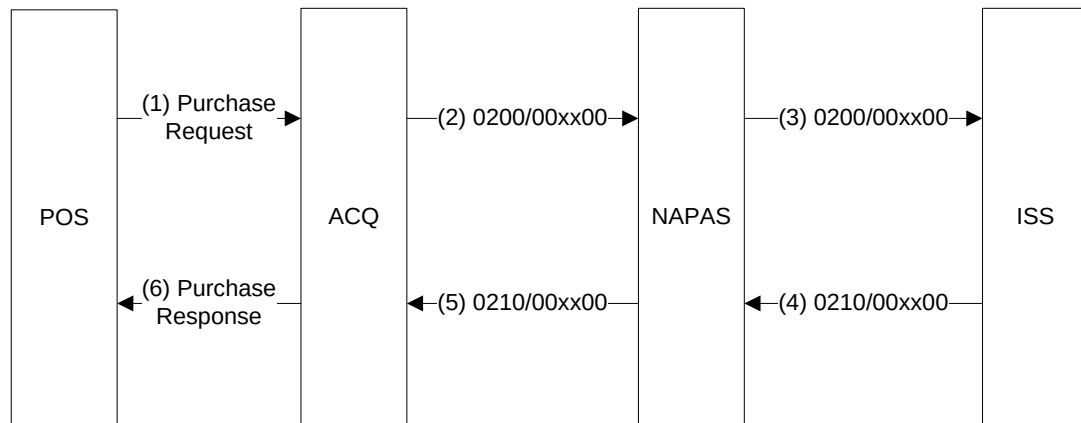


Figure 7. POS purchase transaction flow

Transaction flow detail:

Step	Transaction flow
1	Cardholder selects “ Purchase ” function, swipes card and enters PIN number at a POS POS initiates a purchase request message to ACQ.
2	ACQ sends the purchase request message to NAPAS (Pcode 00xx00).
3	NAPAS forwards the message to ISS.
4	ISS processes and responds a message to NAPAS.
5	NAPAS forwards the responds message to ACQ.
6	ACQ sends the response message to the POS.

4.1.1.2.3 Void

The successful purchase transaction can be cancelled before settling by Void transaction. Void transaction allows customer to cancel all original purchase, customer account will be refunded a full amount of purchase transaction.

At the NAPAS switching system, void transaction is processed according to the mechanism of 0420/0430 message like reversal transaction (refer to more detail information at the section regarding reversal transaction process), meaning ACQ will be responsible to converting internal void message to 0420/0430 message based on NAPAS specification.

NAPAS system does not support to reverse a void transaction.

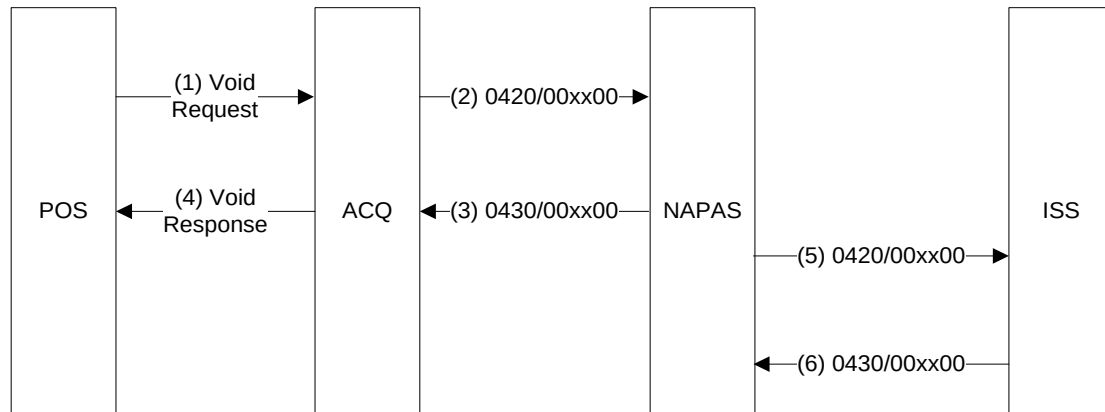


Figure 8. Void transaction flow

Transaction flow detail:

Step	Transaction flow
1	POS initiates a void request message to ACQ.
2	ACQ sends the void request message (0420/00xx00) to NAPAS.
3	NAPAS responds with a message (0430/00xx00) having RC = 00 to ACQ.
4	ACQ returns the void response message to POS.
5	NAPAS forwards the 0420/00xx00 message to ISS.
6	ISS processes and responds with a 0430/00xx00 message having RC = 00 to NAPAS.

4.1.1.2.4 PIN Change

The NAPAS switching allows cardholder of Member Bank who can change the personal identification number (PIN number) via the POS of other bank.

PIN Change message (0200) is initiated from ACQ and sends to NAPAS, based on card range NAPAS will forward this message to ISS.

ISS checks and resends the response message (0210) to ACQ via NAPAS.

In some cases, PIN change was not successful at ACQ but the process of PIN change at ISS was still successful. Therefore, ACQ should notify the customer to re-check after performing PIN change.

PIN change does not have a reversal transaction, if ACQ sends a reversal transaction to NAPAS, NAPAS will reject by a response message having RC <> 00

- New PIN Block value is stored in DE105 (New PIN Block needs to be encrypted with ZPK which like original PIN block in DE52).
- Old PIN Block is stored in DE52.

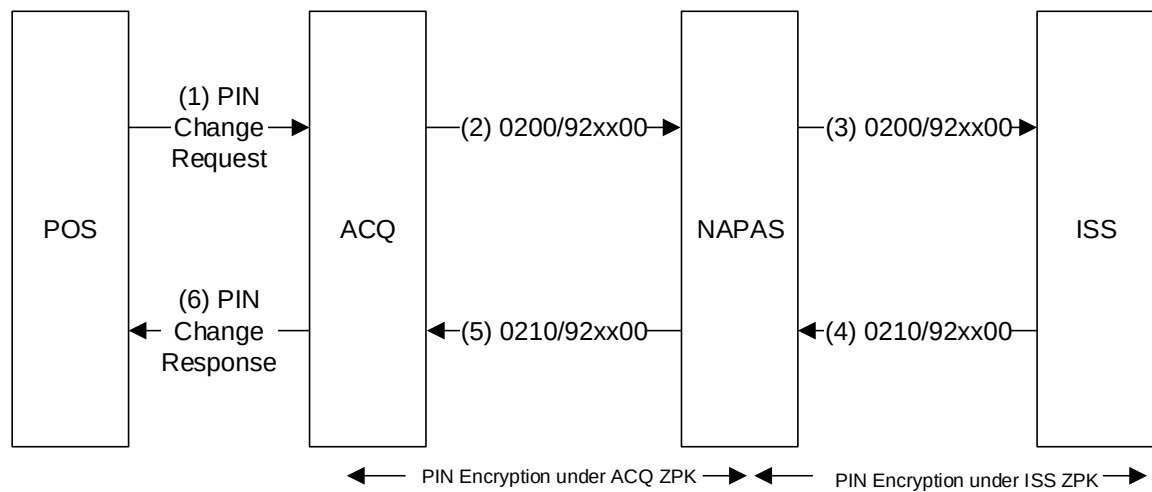


Figure 9. PIN change transaction flow

Transaction flow detail:

Step	Transaction flow
1	Selects "PIN change" function on the screen Cardholder enters old PIN and select Enter button. Cardholder enters new PIN and select Enter button. Cardholder enters new PIN and select Enter button to confirm. If customer enters the PIN 02 times and entered PIN is not the same then POS will prompt customer to re-enter new PIN. POS initiates a PIN change request message to ACQ.
2	ACQ sends the PIN change request message to NAPAS (Pcode 94xx00).
3	NAPAS performs to processing the transaction and forwards it to ISS.
4	ISS checks old PIN of customer before perform to changing new PIN and response a message to NAPAS.
5	NAPAS forwards the response message to ACQ.
6	ACQ sends the response message to POS.

Note: PIN change does not have a reversal transaction, in time-out case, ATM will notify customer to re-check the PIN could change or not:

"PIN change transaction may be processed unsuccessful, please re-check your PIN"

4.1.2 Reversal transaction processing mechanism (0420/0430)

Reversal transaction (0420/0430) is used to reverse transaction of card switching service at the ATM/POS which perform before. Reversal transaction (0420/0430) could be initiated from ACQ or NAPAS.

Card switching service transactions at the ATM/POS have a reversal transaction including as follows:

- Balance inquiry at ATM
- Cash withdrawal at ATM
- Mini-statement at ATM
- Balance inquiry at POS
- Purchase at POS

When receive reversal message (0420) from ACQ, NAPAS responds immediately with a 0430 message having RC = 00 to ACQ.

With 0420/0430 message:

- If NAPAS system finds an origin transaction of 0420 message, NAPAS will forwards a 0420 message to ISS.
- If NAPAS system does not find an origin transaction of 0420 message, NAPAS will not forward a 0420 message to ISS.

In case NAPAS system has not received a 0430 message from ISS, a 0420 reversal message will store in SAF (Store and Forward) of NAPAS system and performs to re-send a 0420 message to ISS. The re-sending a 0420 message times could be configured limitary for each member bank or re-send a 0420 reversal message until NAPAS receives a 0430 response message from ISS.

When ISS receives a 0420 reversal message, ISS will response immediately with 0430 message having RC = 00. The processing of origin transaction at member bank's system could be perform later to increase the system performance.

Note: NAPAS system does not process a reversal transaction which do not have the settlement date same as the current settlement date.

4.1.2.1 Reversal transaction flow (initiated from ACQ)

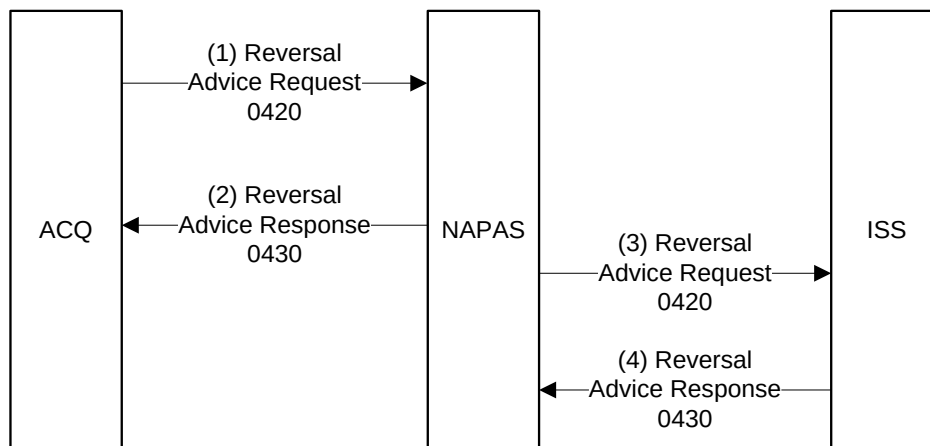


Figure 10. Reversal transaction flow (initiated from ACQ)

Transaction flow detail:

Step	Transaction flow
1	ACQ initiates a reversal message (0420) to NAPAS.

2	NAPAS responds the reversal message (0430) with RC = 00 to ACQ.
3	NAPAS sends the reversal request message (0420) to ISS.
4	ISS responds a message (0430) with RC = 00 to NAPAS.

4.1.2.2 Reversal transaction initial from NAPAS flow

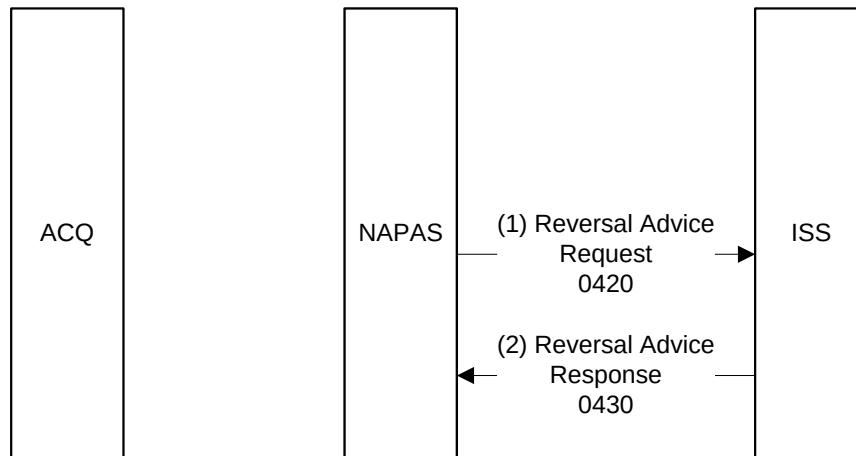


Figure 11. Reversal transaction flow (initiated from NAPAS)

Transaction flow detail:

Step	Transaction flow
1	NAPAS initiates a reversal message (0420) to ISS.
2	ISS responds a reversal message (0430) with RC = 00 to NAPAS.

4.1.3 Exception transaction flow

4.1.3.1 Transaction is rejected at ISS

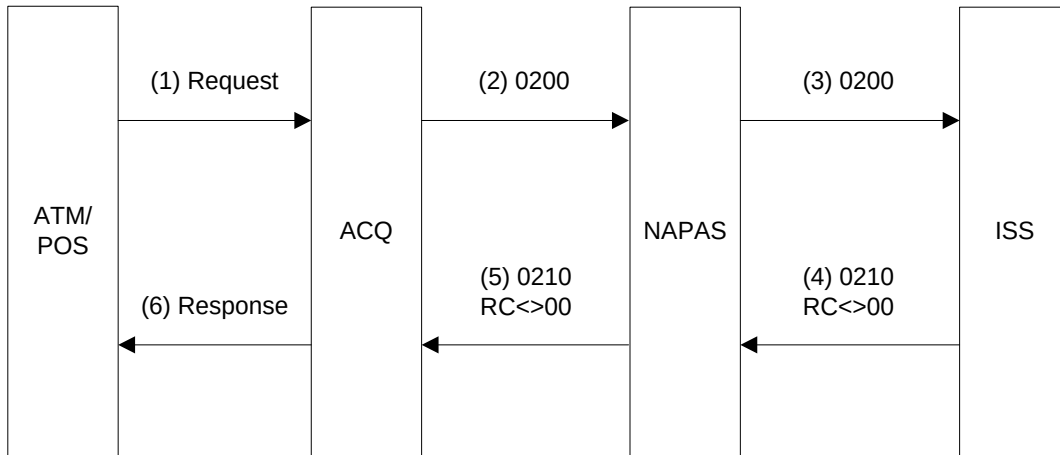


Figure 12. Transaction process flow that is rejected at ISS

Transaction flow detail

Step	Transaction flow
1	ATM/POS initiates a request message and sends it to ACQ.
2	ACQ sends the request message (0200) to NAPAS.
3	NAPAS forwards the message to ISS.
4	ISS processes and refuses to authorize a request transaction by a 0210 response message with RC <> 00 and sends it to NAPAS.
5	NAPAS forwards the response message (0210) with RC <> 00 to ACQ.
6	ACQ sends a response message (reject) to ATM/POS.

4.1.3.2 Rejected Transaction from NAPAS

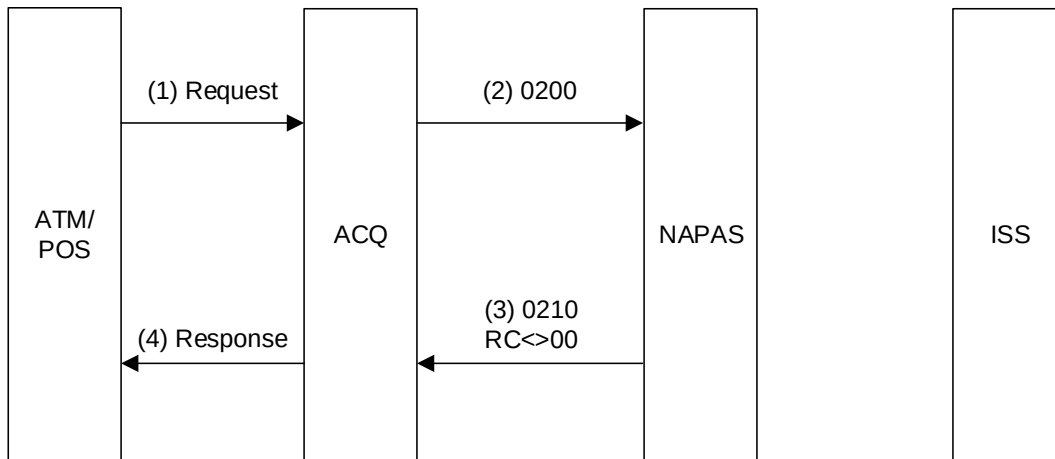


Figure 13. Rejected transaction flow from NAPAS

Transaction flow detail:

Step	Transaction flow
1	ATM/POS initiates a request message and sends it to ACQ.
2	ACQ sends the request message (0200) to NAPAS.
3	NAPAS processes and does not forward the 0200 message (due to reject reason such as wrong format, duplicate message, unavailable ISS, etc.). NAPAS system generates a 0210 response message with RC <> 00 and sends it to ACQ.
4	ACQ sends a response message (reject) to ATM/POS.

4.1.3.3 ACQ sends a reversal transaction for 0210 transaction with RC = 00 at ATM

When ACQ receives a reversal transaction for 0210 message at ATM which RC = 00, usually due to following causes:

- With cash withdrawal: ATM is disconnected to ACQ and not received a refund request; ATM is malfunction and does not have the ability to dispense money, etc.
- With transfer transaction: ATM is disconnected with ACQ, etc....

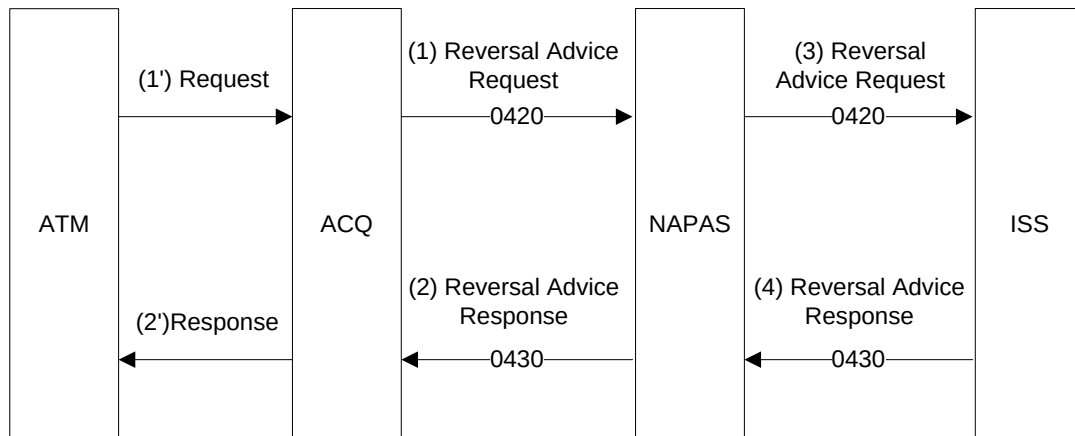


Figure 14. Reversal transaction from ACQ for 0210 transaction with RC =00 at ATM

Transaction flow detail:

Step	Transaction flow
1'	ACQ receives the processing result from ATM which was not successful or too time-out to get feedback from ATM.
1	ACQ creates a reversal message (0420) and sends it to NAPAS.
2	NAPAS responds a reversal message (0430) with RC = 00 to ACQ.
2'	ACQ sends a reversal response message (0430) to ATM.
3	NAPAS sends the reversal message (0420) to ISS.
4	ISS responds a message (0430) with RC = 00 to NAPAS.

Note: If ISS responds a reversal message (0430) with RC <> 00, NAPAS still considers this transaction as a successful reversal.

4.1.3.4 ACQ sends a reversal transaction for 0210 transaction with RC = 00 at POS

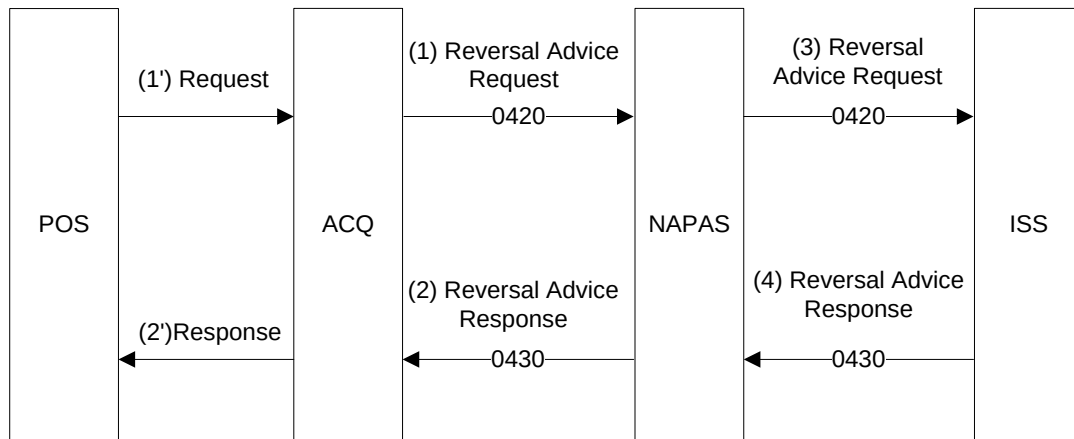


Figure 15. Reversal transaction from ACQ for 0210 transaction with RC =00 at POS

Transaction flow detail:

Step	Transaction flow
1'	POS initiates a reversal message (0420) to ACQ.
1	ACQ sends a reversal message (0420) to NAPAS.
2	NAPAS responds a reversal message (0430) with RC = 00 to ACQ.
2'	ACQ sends the reversal response message to POS.
3	NAPAS sends the reversal message (0420) to ISS.
4	ISS responds a message (0430) with RC = 00 to NAPAS.

Note: if ISS responds a reversal message (0430) with RC <> 00, NAPAS still considers this transaction as a successful reversal.

4.1.3.5 ACQ send a reversal transaction for 0210 transaction with RC <> 00

NAPAS confirms that a 0210 transaction with RC <> 00 and still receives a reversal transaction from ACQ.

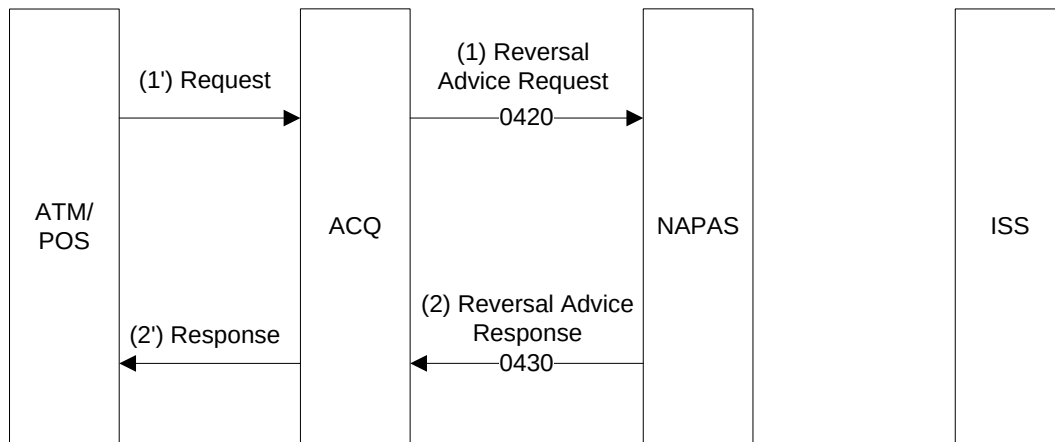


Figure 16. ACQ sends a reversal transaction for 0210 transaction with RC <> 00

Transaction flow detail:

Step	Transaction flow
1'	ATM/POS initiates a reversal message (0420) and sends it to ACQ.
1	ACQ receives a reversal message (0420) and sends it to NAPAS.
2	NAPAS responds a reversal message (0430) with RC = 00 to ACQ.
2'	ACQ sends a reversal response message to ATM.
	In this case, NAPAS does not send a reversal message (0420) to ISS.

4.1.3.6 Time-out transaction at NAPAS

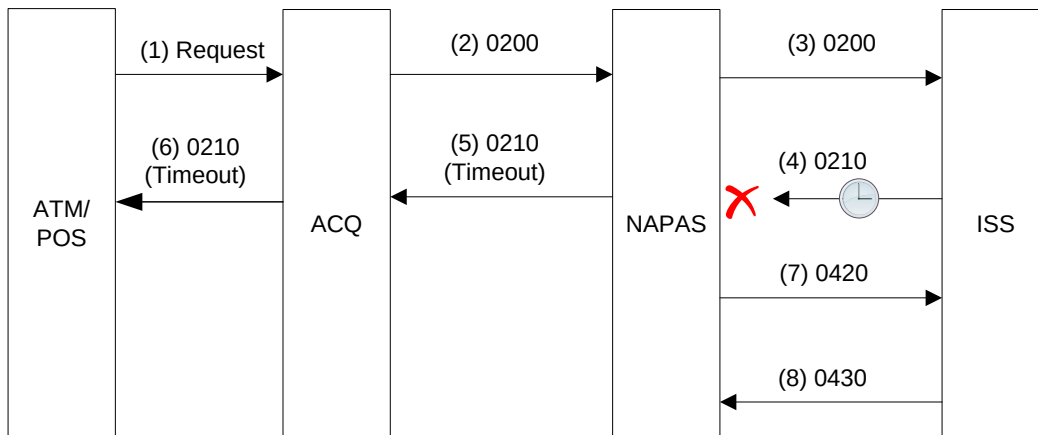


Figure 17. Time-out transaction flow at NAPAS

Transaction flow detail:

Step	Transaction flow
1	ATM/POS initiates a request message and sends it to ACQ.
2	ACQ sends a request message (0200) to NAPAS.
3	NAPAS forwards a message (0200) to ISS.
4	After time-out period, NAPAS does not receive the response message (0210) from ISS.
5	NAPAS initiates a response message (0210) with RC as time-out and sends it to ACQ.
6	ACQ sends a response message with response code as time-out to ATM/POS.
7	Simultaneously with step 5, NAPAS initiates a reversal message (0420) and sends it to ISS.
8	ISS responds a reversal message (0430) with RC = 00.

If ACQ receives the response message (0210) with response code as time-out from NAPAS then ACQ does not have to send a reversal message (0420) to NAPAS.

4.1.3.7 Time-out transaction at ACQ

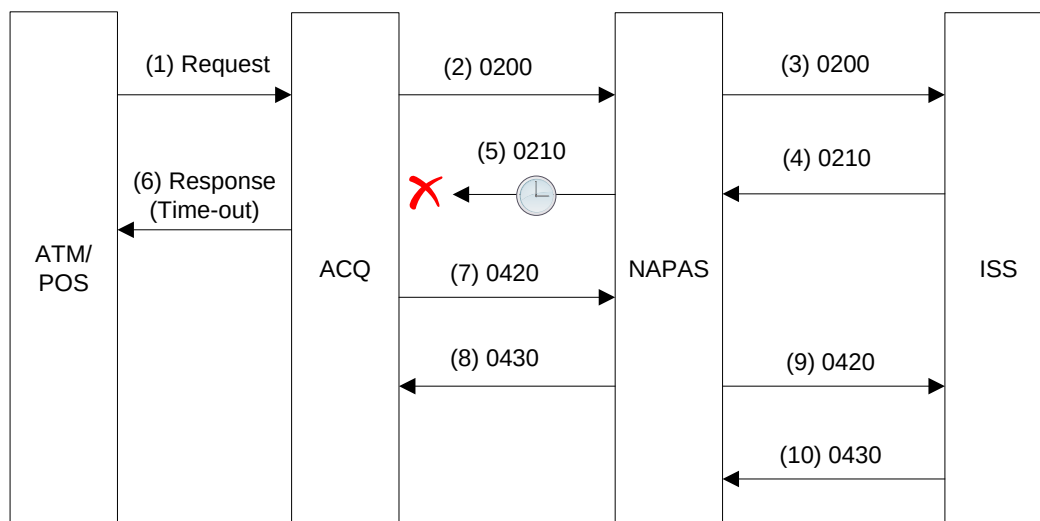


Figure 18. Time-out transaction flow at ACQ

Transaction flow detail:

Step	Transaction flow
1	ATM/POS initiates a request message and sends it to ACQ.
2	ACQ sends the request message (0200) to NAPAS.
3	NAPAS forwards the message (0200) to ISS.
4	ISS processes and responds the message (0210) to NAPAS.
5	After time-out period, ACQ have not received the response message (0210) from NAPAS.
6	ACQ initiates a response message with response code as time-out and sends it to ATM/POS.
7	Simultaneously with step 6, ACQ initiates a reversal message (0420) and sends it to NAPAS.
8	NAPAS responds a reversal message (0430) with RC = 00 to ACQ.
9	NAPAS forwards the reversal message (0420) to ISS.
10	ISS responds the reversal message (0430) with RC = 00 to NAPAS.

4.1.3.8 Late response transaction at ISS

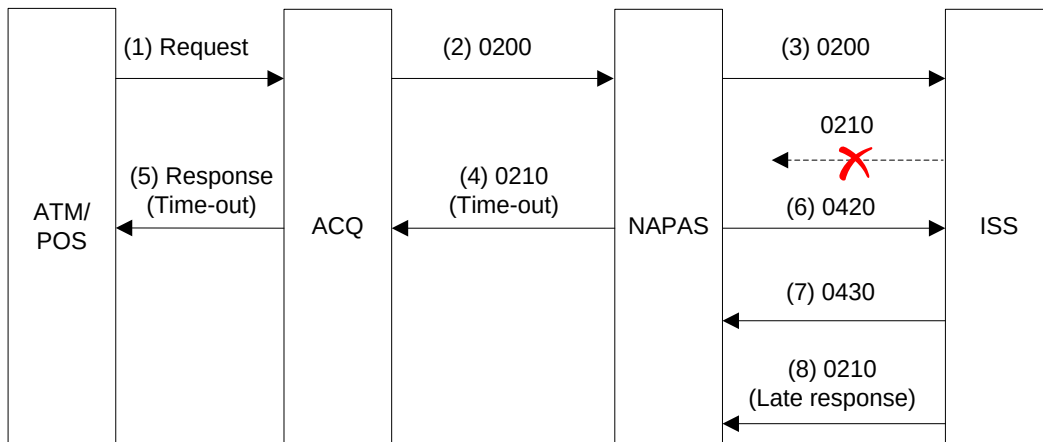


Figure 19. Late response transaction flow at ISS

Transaction flow detail:

Step	Transaction flow
1	ATM/POS initiates a request message and sends it to ACQ.
2	ACQ sends the request message (0200) to NAPAS.
3	NAPAS forwards the message (0200) to ISS.
4	After time-out period, NAPAS have not received a response message (0210) from ISS, NAPAS initiates a response message (0210) with response code as time-out and sends it to ACQ.
5	ACQ sends the response message (time-out) to ATM/POS.
6	Simultaneously with step 4, NAPAS initiates a reversal message (0420) and sends it to ISS.
7	ISS responds a reversal message (0430) with RC = 00 to NAPAS.
8	After a time-out period, NAPAS have received the response message (0210) with RC = 00 from ISS, this transaction is considered as late response transaction and is rejected by NAPAS (NAPAS does not send it to ACQ)

If ACQ receives a response message (0210) with response code as time-out from NAPAS then ACQ must not send a reversal message (0420) to NAPAS.

4.1.3.9 Time-out reversal transaction at ISS

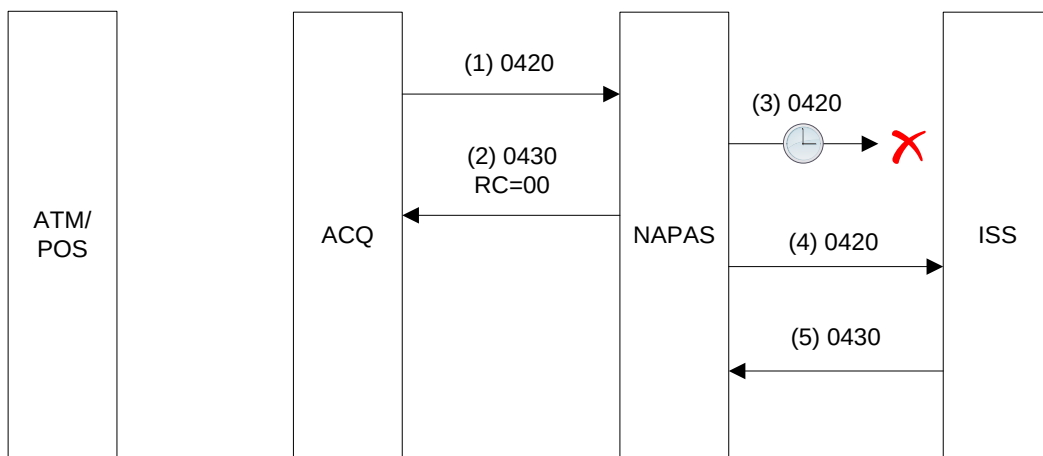


Figure 20. Time-out reversal transaction flow at ISS

Transaction flow detail:

Step	Transaction flow
1	ACQ sends a request message (0420) to NAPAS.
2	NAPAS responds a reversal message (0430) with RC = 00 to ACQ.
3	NAPAS forwards the reversal message (0420) to ISS, but ISS have not received the reversal message (0420) (such as errors caused by disconnection line, etc.)
4	After time-out period, NAPAS continue to resending reversal message (0420) to ISS.
5	ISS responds a reversal message (0420) which is duplicated by the message (0430) with RC = 00 to NAPAS.

4.1.3.10 Duplicate reversal transaction at NAPAS

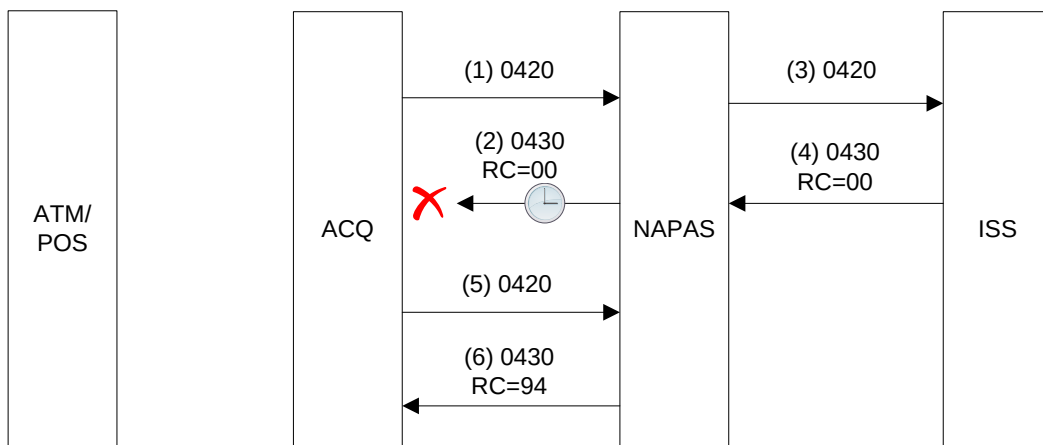


Figure 21. Duplicate reversal transaction flow at NAPAS

Transaction flow detail:

Step	Transaction flow
1	ACQ sends a reversal message (0420) to NAPAS.
2	NAPAS responds a reversal message (0430) with RC = 00 to ACQ, but ACQ have not received response message (0430) (such as errors caused by disconnection line, etc.).
3	NAPAS forwards the reversal message (0420) to ISS.
4	ISS responds a reversal message (0430) with RC = 00 to NAPAS.

5	After time-out period, ACQ resends the reversal message (0420) to NAPAS
6	NAPAS responds a reversal message (0420) which is duplicated by the message (0430) with RC = 94 to ACQ. Duplicate reversal messages will not be forwarded to ISS.

When ACQ have received a response reversal message (0430) with RC = 94 (duplicate reversal transaction), ACQ considers it as successful transaction.

4.1.3.11 Duplicate reversal transaction at ISS

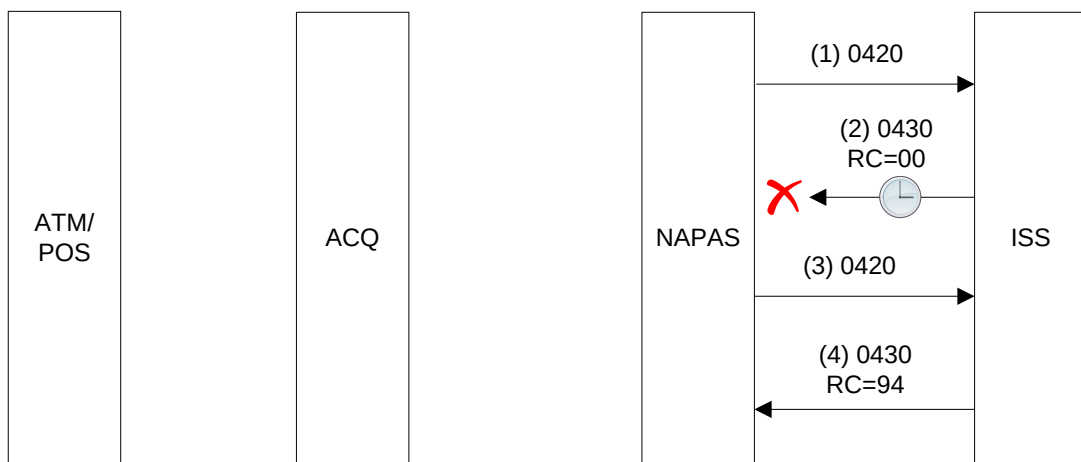


Figure 22. Duplicate reversal transaction flow at ISS

Transaction flow detail:

Step	Transaction flow
1	NAPAS sends a reversal message (0420) to ISS.
2	ISS processes and responds a reversal message (0430) with RC = 00 to NAPAS, but NAPAS have not received a response message (0430) from ISS (such as errors caused by disconnection line, etc.)
3	After time-out period, NAPAS continues to resending the reversal message (0420) to ISS.
4	ISS responds a reversal message (0420) which is duplicated by the message (0430) with RC = 94 (duplicate reversal transaction) to NAPAS.

When NAPAS receives the reversal response message (0430) with **RC = 94** (duplicate reversal transaction), NAPAS considers it as a successful transaction.

4.2 Online switching transaction (Ecom)

4.2.1 Successful processing flow

NAPAS provides online payment solutions for Banks and Payment gateway which connect via NAPAS's switching system (Ecom), support to pay using customer card.

4.2.1.1 Regular payment transaction flow (Ecom Purchase)

Online switching service (Ecom) includes 03 transaction types as follows (card information, OTP number is inputted at Payment gateway website):

- Verify card information (Verify Card)
- Verify OTP information (Verify OTP)
- Payment (Ecom purchase)

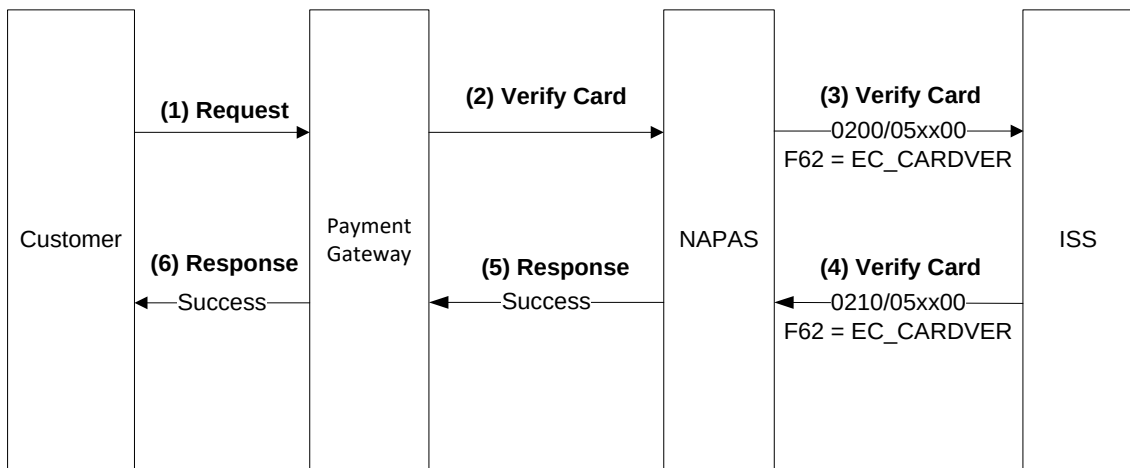


Figure 23. Card information verification transaction flow (Verify card)

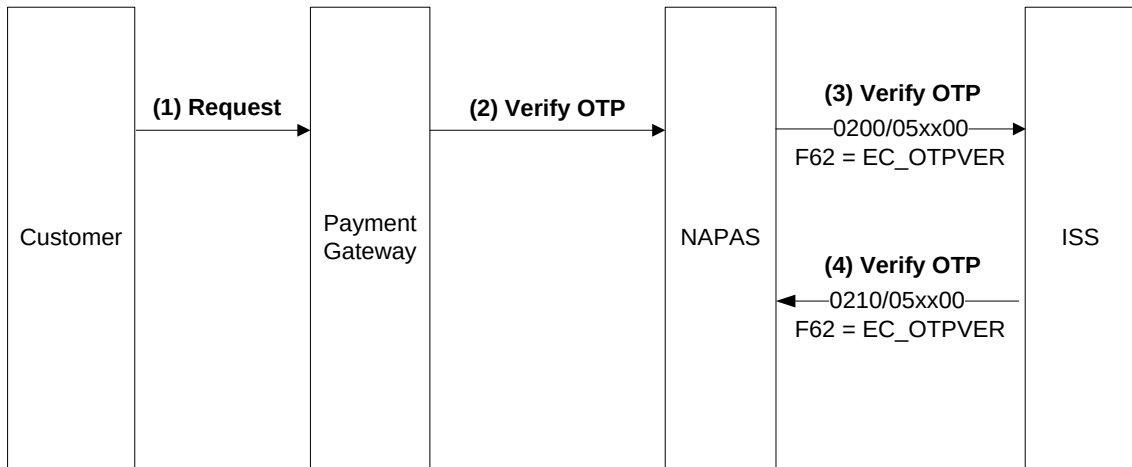


Figure 24. OTP verification transaction flow (Verify OTP)

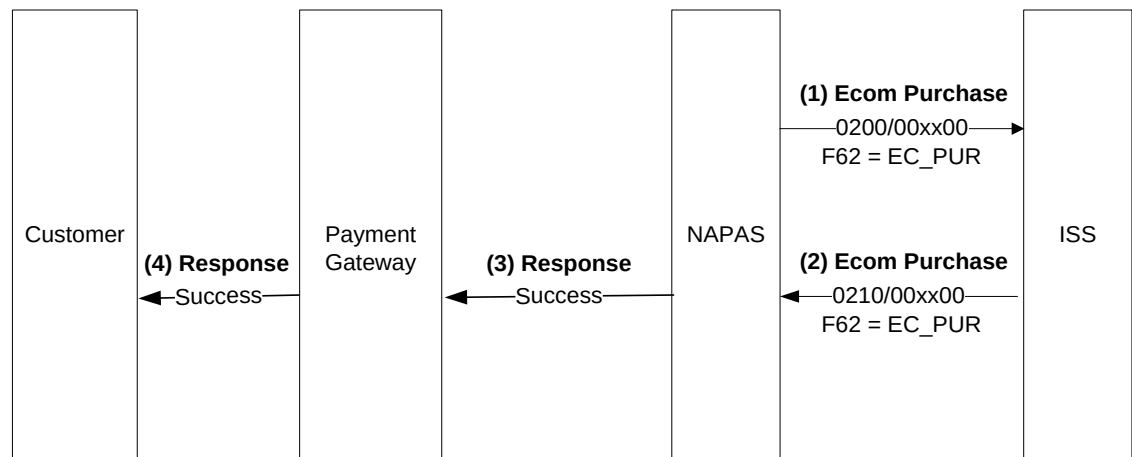


Figure 25. Payment transaction flow (Ecom purchase)

Transaction flow detail:

Step	Transaction flow
Card information verification transaction (Verify card)	
1	Customer begins to pay for the purchase on web/interface of payment gateway. Customer enters a cardholder information on web/interface of payment gateway (Verify card).
2	Payment gateway sends a cardholder information verification request to NAPAS.
3	NAPAS have received a cardholder information verification request and sends to ISS to perform processing.

4	ISS authorizes customer's card and account information. If card and account are valid, ISS generates an OTP and sends it to customer, and responds a successful customer authorization result to NAPAS.
5	NAPAS resends result of successful cardholder information verification to Payment gateway.
6	Payment gateway interface goes to the OTP verification screen.
OTP verification transaction (Verify OTP)	
1	The customer enters an OTP which ISS sends on Payment gateway interface.
2	Payment gateway sends an OTP verification request (Verify OTP) to NAPAS.
3	NAPAS receives the OTP verification request (Verify OTP) and sends it to ISS to processing the request.
4	ISS verifies an OTP successful and sends a verification result to NAPAS.
Payment transaction (Ecom purchase)	
1	After receiving successful OTP verification result, NAPAS initiates a payment transaction (Ecom Purchase) and sends it to ISS to perform processing request.
2	ISS processes a payment request (debits customer card account) and sends a successful payment result to NAPAS.
3	NAPAS sends the successful payment result to Payment Gateway.
4	Payment gateway informs a successful transaction to customer and finishes online payment transaction flow.

Note: in case customer enters payment information on NAPAS 's payment gateway interface, cardholder and OTP verification information will show step by step on payment gateway provided by NAPAS. The system only returns latest payment transaction to Payment gateway.

4.2.1.2 Fast payment transaction flow (Ecom Fast Purchase)

Beside above Ecom service includes 03 transaction types, NAPAS also supports to debit customer account by a single transaction (Ecom Fast Purchase):

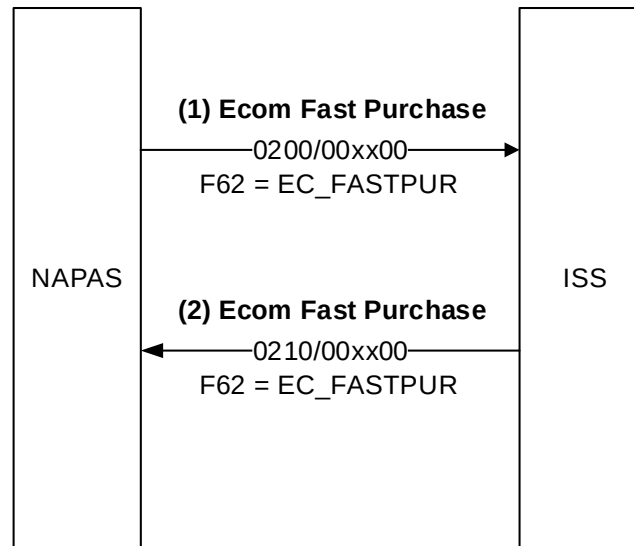


Figure 26. Ecom fast payment processing flow (Ecom Fast Purchase)

Step	Transaction flow
Fast payment transaction (Ecom Fast Purchase)	
1	NAPAS initiates a payment transaction (Ecom Fast Purchase) and sends it to ISS to processing request.
2	ISS verifies customer's card/account information and payment authentication information. If these information are valid, ISS will process the payment request (debits customer's card account) and sends the successful result to NAPAS. NAPAS receives a successful transaction and finishes transaction processing.

4.2.2 Reversal transition processing mechanism (0420/0430)

A reversal transaction (0420/0430) is used to reverse a payment transaction (Ecom Purchase) and a fast payment transaction (Ecom Fast Purchase) which is performed before. The reversal transaction (0420/0430) is initiated from NAPAS.

When ISS receives a reversal transaction (0420), ISS will response immediately by a message (0430) with RC = 00. The processing for the original transaction at member bank's system could be perform later to increase system performance.

Note: NAPAS system does not process a reversal transaction which have not got settlement date same as the current settlement date.

Reversal transaction processing flow is performed as follows:

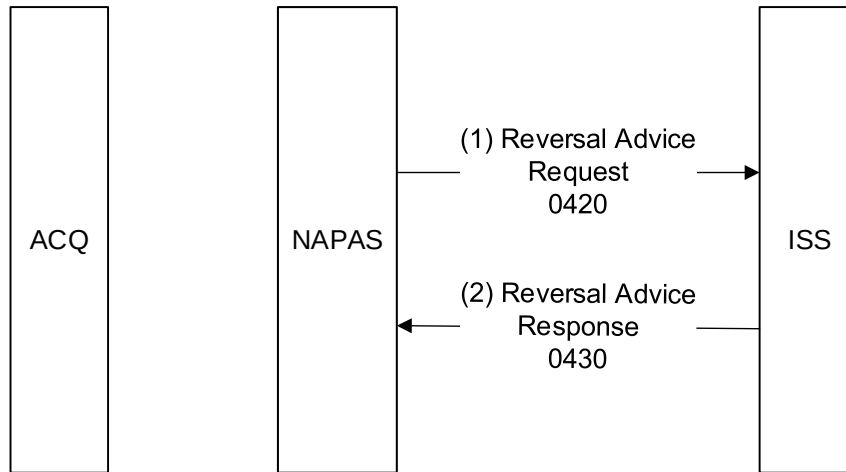


Figure 27. Reversal transaction flow

Transaction flow detail:

Step	Transaction flow
1	NAPAS initiates a reversal message (0420) and sends it to ISS.
2	ISS responds a reversal message (0430) and sends it to NAPAS.

4.2.3 Exception transaction flow

4.2.3.1 Time-out cardholder information authentication between NAPAS and ISS

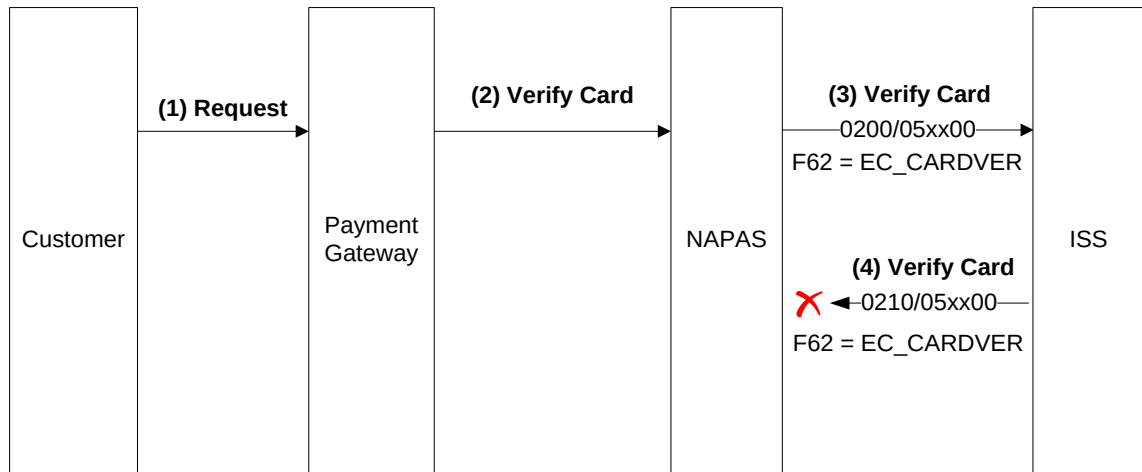


Figure 28. Time-out cardholder information authentication flow between NAPAS and ISS

Transaction flow detail:

Step	Transaction flow
1	Customer begins to pay for the purchase on web/interface of Payment gateway. The customer enters a card holder information on payment gateway interface (Verify card).
2	Payment gateway sends a card holder information verification request to NAPAS.
3	NAPAS receives a cardholder information verification request and sends it to ISS to processing.
4	ISS resends the response message to NAPAS but NAPAS have not received. The transaction is marked as time-out. Finish processing online payment transaction.

4.2.3.2 Time-out cardholder information authentication transaction between NAPAS and Payment gateway

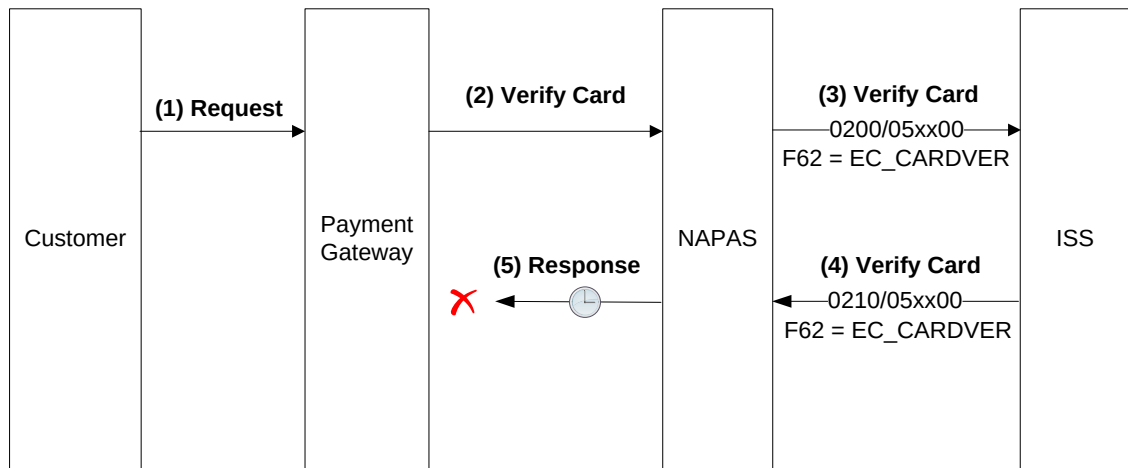


Figure 29. Time-out Cardholder information authentication transaction flow between NAPAS and Payment gateway

Transaction flow detail:

Step	Transaction flow
1	Customer begins to pay for the purchase on web/interface of Payment gateway. The customer enters a card holder information on payment gateway interface (Verify card).
2	Payment gateway sends a card holder information verification request to NAPAS.
3	NAPAS have received cardholder information verification request and sends it to ISS to processing.
4	ISS responds cardholder information verification result to NAPAS.
5	NAPAS sends an authenticated response transaction to Payment gateway but they does not receive the message. Finishes processing the online transaction. Dispute transaction processing will follow with exception transaction processing regulation for Payment Gateway.

4.2.3.3 Time-out OTP authentication transaction between NAPAS and ISS

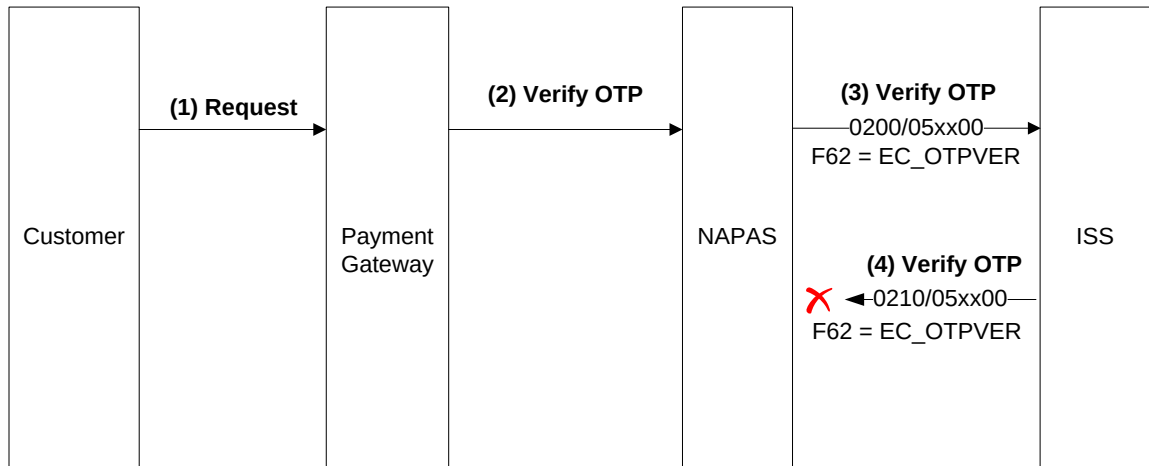


Figure 30. Time-out OTP authentication transaction flow between NAPAS and ISS

Transaction flow detail:

Step	Transaction flow
1	The customer enters OTP information from ISS on payment interface.
2	Payment Gateway sends an OTP verification request (Verify OTP) to NAPAS.
3	NAPAS receives an OTP verification request and sends to ISS to process request.
4	ISS resends a response message to NAPAS but NAPAS does not receive. The transaction is marked as time-out. Finish processing online payment transaction.

4.2.3.4 Time-out OTP authentication transaction between NAPAS and Payment gateway

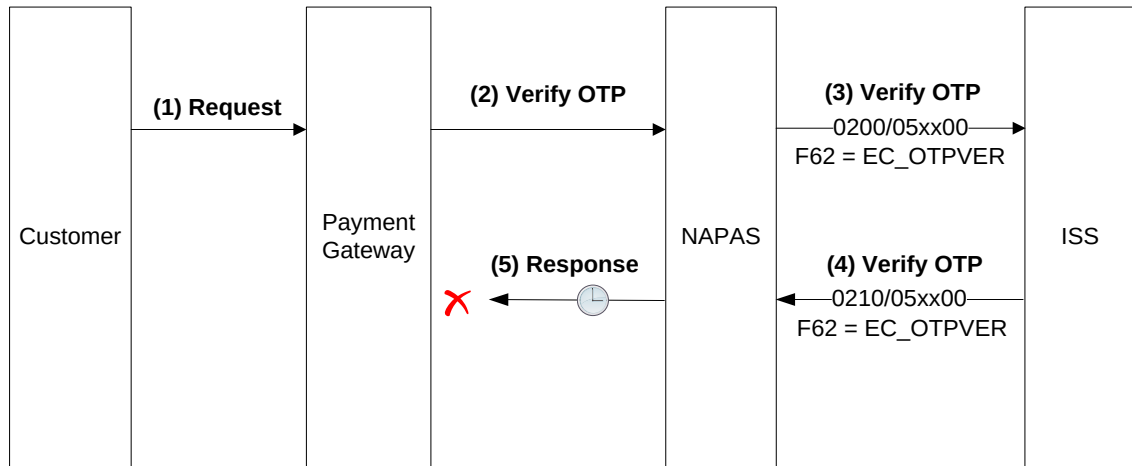


Figure 31. Time-out OTP authentication transaction flow between NAPAS and Payment gateway

Transaction flow detail:

Step	Transaction flow
1	The customer enters an OTP which ISS sent to on payment interface.
2	Payment gateway sends OTP verification request (Verify OTP) to NAPAS.
3	NAPAS receives an OTP verification request and sends to ISS to processing request
4	ISS authenticates an OTP successfully and sends it to NAPAS.
5	NAPAS sends authenticated response transaction to Payment gateway but they does not receive. Finish processing online payment transaction.

4.2.3.5 Time-out purchase transaction between NAPAS and ISS

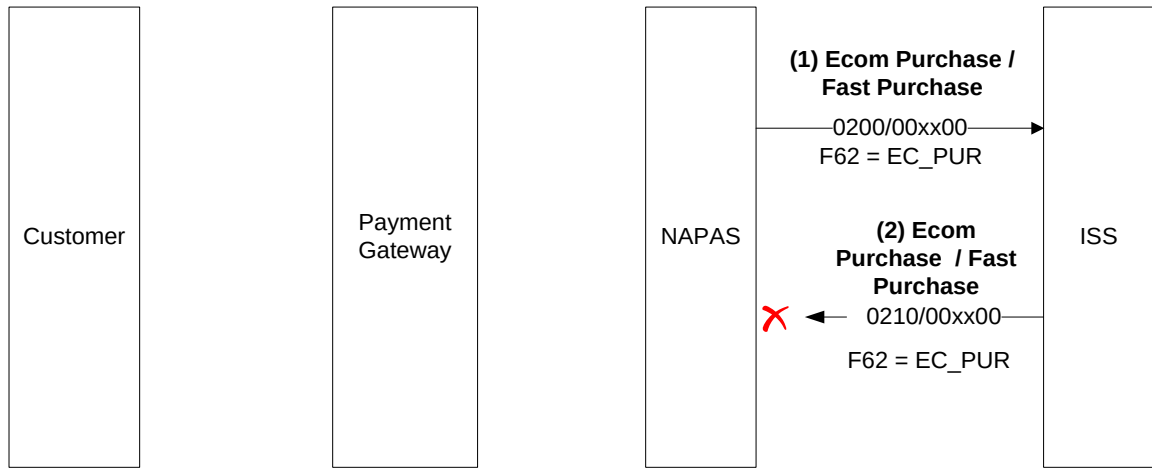


Figure 32. Time-out purchase transaction flow between NAPAS and ISS

Transaction flow detail:

Step	Transaction flow
1	NAPAS initiates a payment transaction (Ecom Purchase) / a fast payment transaction (Ecom Fast Purchase) and sends it to ISS to perform processing request.
2	ISS resends a response message to NAPAS but NAPAS does not receive. The transaction is marked as time-out. Finish processing the online payment.

4.2.3.6 Time-out purchase transaction between NAPAS and Payment gateway

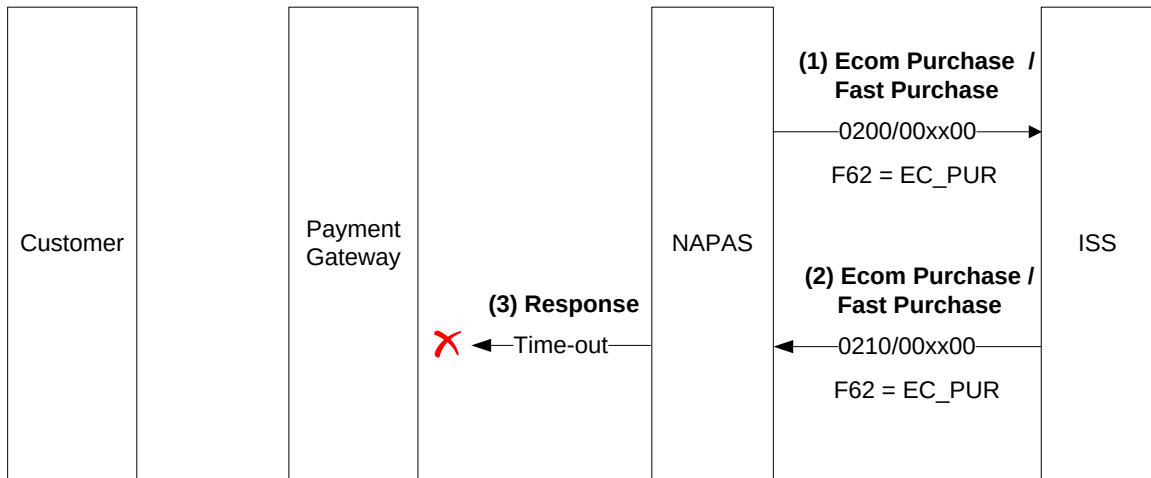


Figure 33. Time-out purchase transaction flow between NAPAS and Merchant

Transaction flow detail:

Step	Transaction flow
1	NAPAS initiates a payment transaction (Ecom purchase) and a fast payment transaction (Ecom Fast Purchase) and sends it to ISS to perform processing request.
2	ISS processes and responds the request message to NAPAS.
3	NAPAS sends an authenticated response transaction to payment gateway but they does not receive. Finish processing a payment transaction. Dispute transaction processing will follow with exception transaction processing regulation for Payment Gateway.

4.2.3.7 Expired OTP

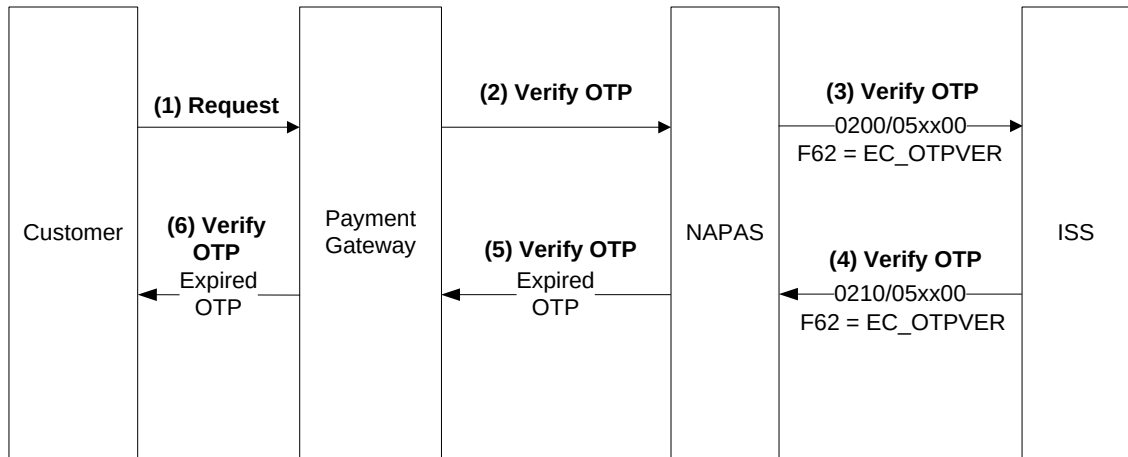


Figure 34. Expired OTP transaction flow

Transaction flow detail:

Step	Transaction flow
1	The customer enters OTP which ISS sent to customer on payment interface of Payment Gateway.
2	Payment gateway sends OTP verification request (verify OTP) to NAPAS.
3	NAPAS receives an OTP verification request and sends it to ISS to processing request.
4	ISS checks an OTP verification request but OTP is expired (Time-out) then responds back to NAPAS with time-out OTP status.
5	NAPAS sends a response notification of expired OTP to Payment gateway.
6	Payment gateway resends a time-out error notification to customer. ISS resends new OTP to the customer and the customer makes the transaction again. System allows to entering OTP maximum 03 times in a session.

4.2.3.8 Unsuccessful cardholder information authentication

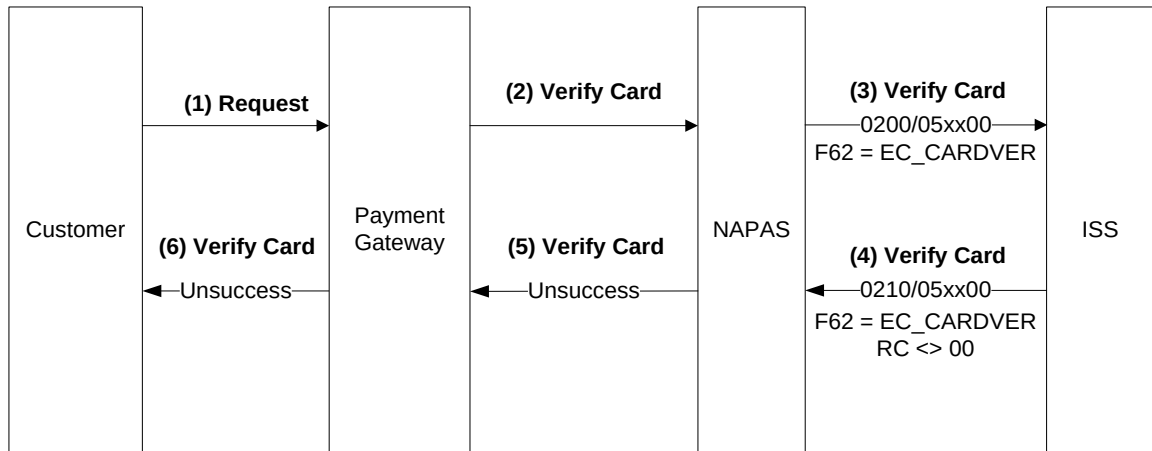


Figure 35. Unsuccessful cardholder information authentication transaction flow

Transaction flow detail:

Step	Transaction flow
1	Customer enters cardholder information on payment interface of Payment Gateway.
2	Payment gateway sends a cardholder information authentication to NAPAS.
3	NAPAS sends the request to ISS.
4	ISS checks cardholder information authentication request and returns an unsuccessful cardholder information authentication.
5	NAPAS receives unsuccessful cardholder information authentication from ISS and forwards it to Payment Gateway.
6	Payment Gateway resends unsuccessful transaction to the customer (refer more detail these RC <> 00 at section Ecom response code table).

4.2.3.9 Unsuccessful OTP authentication

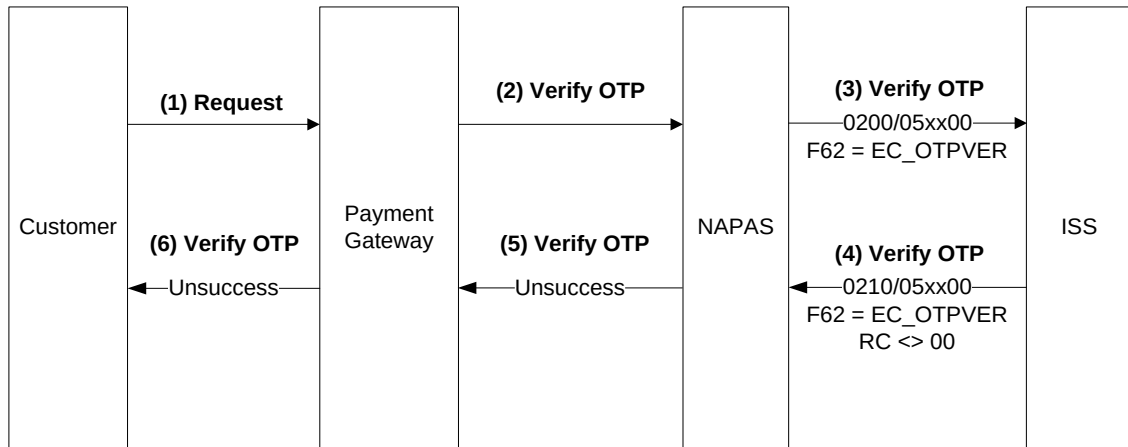


Figure 36. Unsuccessful OTP authentication flow

Transaction flow detail:

Step	Transaction flow
1	Customer enters an OTP information on payment interface of Payment Gateway.
2	Payment gateway sends an OTP authentication request to NAPAS.
3	NAPAS sends the request to ISS.
4	ISS checks an OTP authentication request and returns an unsuccessful OTP authentication.
5	NAPAS sends an unsuccessful OTP authentication response transaction to Payment gateway.
6	Payment gateway informs an unsuccessful transaction to the customer. Refer more detail these RC <> 00 at section Ecom response code table.

4.2.3.10 Unsuccessful payment transaction

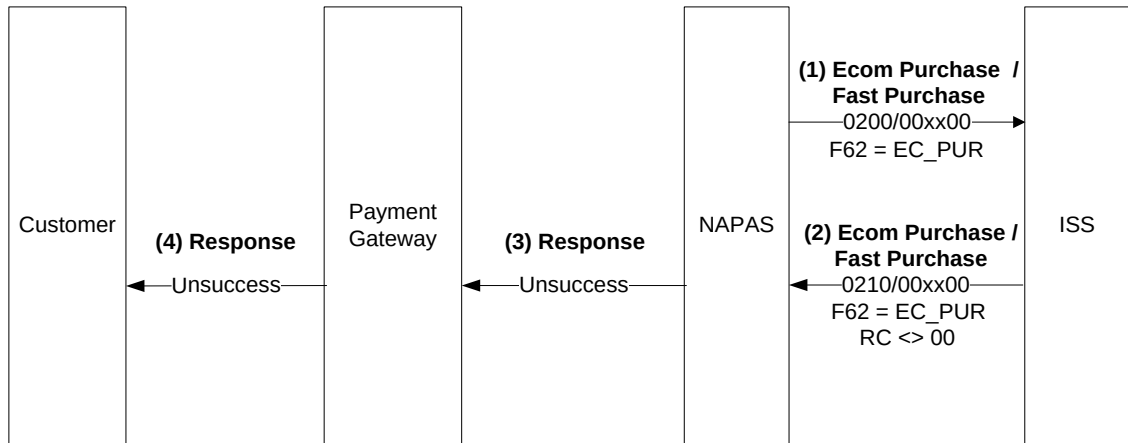


Figure 37. Unsuccessful payment transaction flow

Transaction flow detail:

Step	Transaction flow
1	NAPAS sends an Ecom Purchase / an Ecom Fast Purchase to ISS.
2	ISS processes a request and returns unsuccessful payment transaction error (Ecom purchase) and sends it to NAPAS.
3	NAPAS sends an unsuccessful notification to Payment gateway.
4	Payment gateway resends an unsuccessful transaction to the customer. Detail with RC <> 00 cases, refer to response code table for Ecom service.

4.2.3.11 Time-out reversal transaction is not sent to ISS

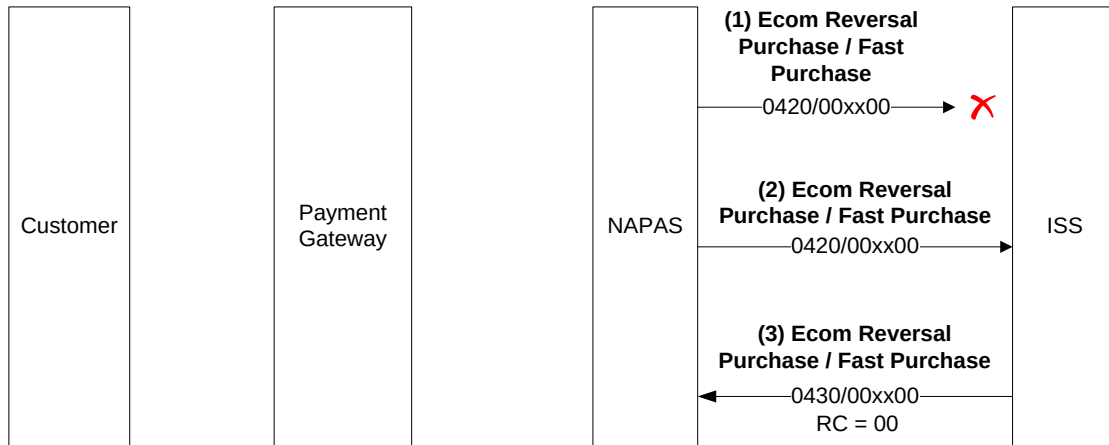


Figure 38. Time-out reversal transaction flow is not sent to ISS

Transaction flow detail:

Step	Transaction flow
1	NAPAS sends a reversal message (0420) to ISS.
2	ISS does not receive reversal message (0420) which NAPAS sent.
3	After time-out period, NAPAS continue sending the reversal message (0420) to ISS.
4	ISS responds a reversal message (0420) by the message (0430) with RC = 00 to NAPAS.

4.2.3.12 Duplicate reversal transaction at ISS

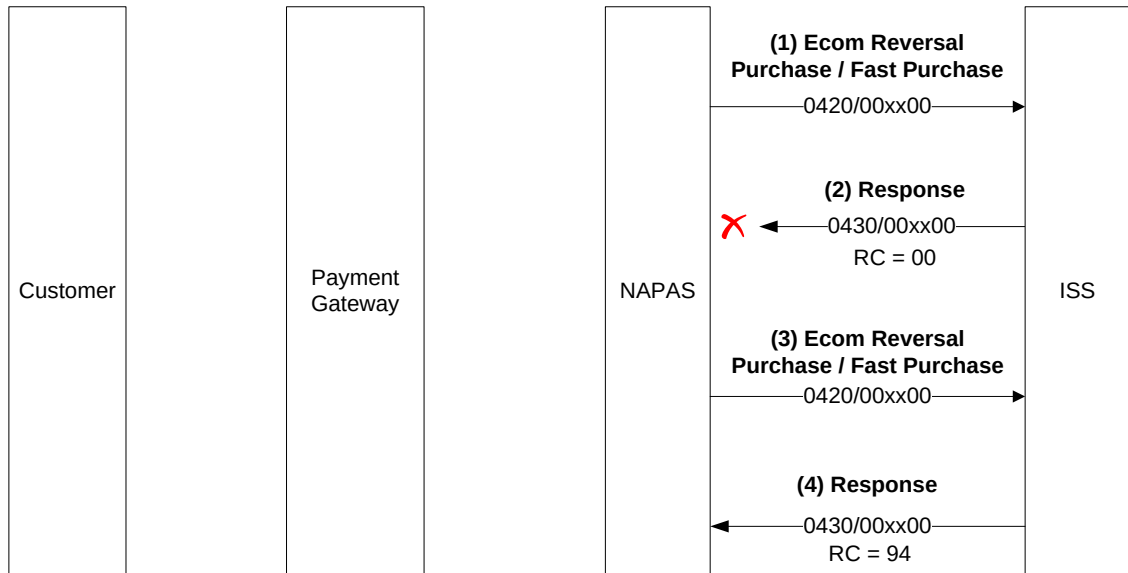


Figure 39. Duplicate reversal transaction flow at ISS

Transaction flow detail:

Step	Transaction flow
1	NAPAS sends a reversal message (0420) to ISS.
2	ISS processes and responds a reversal message (0430) with RC = 00 to NAPAS but NAPAS have not received the response message (0430) from ISS (such as errors caused by disconnect line, etc.).
3	After time-out period, NAPAS continues resending the reversal message (0420) to ISS.
4	ISS responds a reversal message (0420) which is duplicated by the message (0430) with RC = 94 (duplicate reversal transaction) to NAPAS.

When NAPAS have received a reversal response message (0430) with **RC = 94** (duplicate reversal transaction), NAPAS considers it as successful transaction.

4.3 Inter-Bank Fund Transfer 24/7 (IBFT)

4.3.1 Successful transaction flow

NAPAS provide inter-bank fund transfer 24/7 service for customers to transfer money in Interbank (Inter-Bank Fund Transfer – IBFT) on many different payment channels such as ATM, Internet banking, Mobile banking, Counters, etc.).

IBFT transaction is online interbank transfer transaction that debit and credit customer account real time. Money is transferred from an account to another account in real time, is made by debit card. IBFT service support interbank transfer transactions at the ATM of the bank where the customers open their accounts (ACQ = ISS <> BNB).

- MAC is required value for the IBFT transactions.
- Amount limit of IBFT transactions is determined by the card issuing bank.
- Support transfer money using card or account number.
- The distinction between card number and account number is made by the beneficiary bank.

IBFT service include two types of transactions:

- Inquiry card holder/beneficiary account information (IBFT Inquiry).
- Transfer money to the beneficiary card / account (IBFT Deposit).

IBFT transaction process:

- IBFT transaction is processed only if the information of card holder/beneficiary account query transaction and transfer transaction to the cardholder/account are success.
- IBFT transaction will not be performed as if the following situations occur:
 - Verify beneficiary cardholder / account information (IBFT Inquiry) is not successful
 - Customers choose the CANCEL command on the beneficiary cardholder / account information authentication screen.
 - Beneficiary cardholder / account information transaction is not successful such as: IBFT inquiry is refused; invalid account, not active; your account is not active; the transaction was time out, etc.

- When BNB returns the authentication transaction information of card holder/beneficiary account successfully but did not have the name of the beneficiary.

The flow of the overall transaction processing including 02 transaction processing flows are IBFT Inquiry and IBFT Deposit:

The following flow describes 02 transactions of IBFT:

- Beneficiary card/account inquiry transaction process flow:

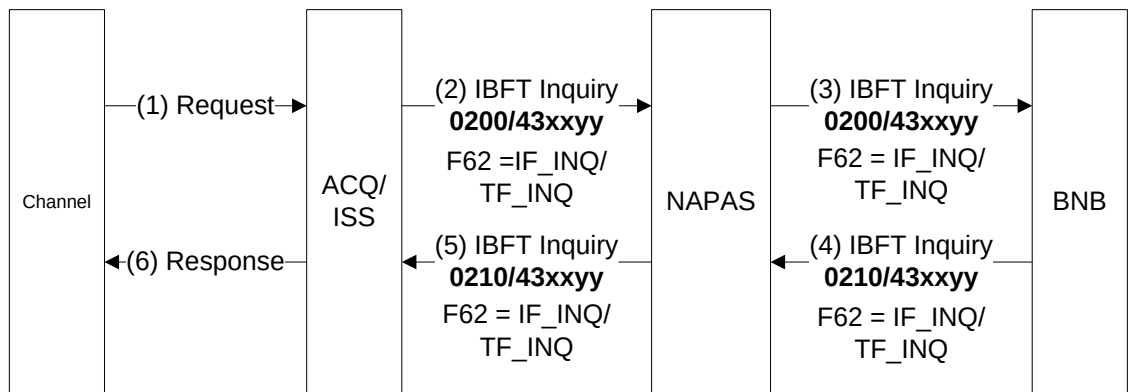


Figure 40. Beneficiary card/account inquiry transaction process flow (IBFT Inquiry flow)

Transaction flow detail:

Step	Transaction flow
1	Customers accesses the ACQ/ISS channel to authenticate beneficiary card holder/ account information.
	Choose one of the interbank funds transfer function as follows: • Transfer money to the card number. • Transfer money to the account number.
	Customers chooses the beneficiary Bank (In case transfer money to the account number) and proceed: • Enters beneficiary card / account number. • Enter the amount to transfer. • Other transaction information (if available).
2	ACQ/ISS sends a beneficiary cardholder/account information inquiry

	request (IBFT Inquiry) to NAPAS.
3	NAPAS receives the request and bases on transaction information to routing the transaction to the beneficiary bank (BNB).
4	BNB will verify the beneficiary information and answers a successful verification transaction, include beneficiary information (if available).
5	NAPAS receives result answer and responds the query result to ACQ/ISS.
6	ACQ/ISS displays successful beneficiary cardholder/account information query and shows beneficiary cardholder (if any).

- Transfer money to beneficiary card/account transaction process flow

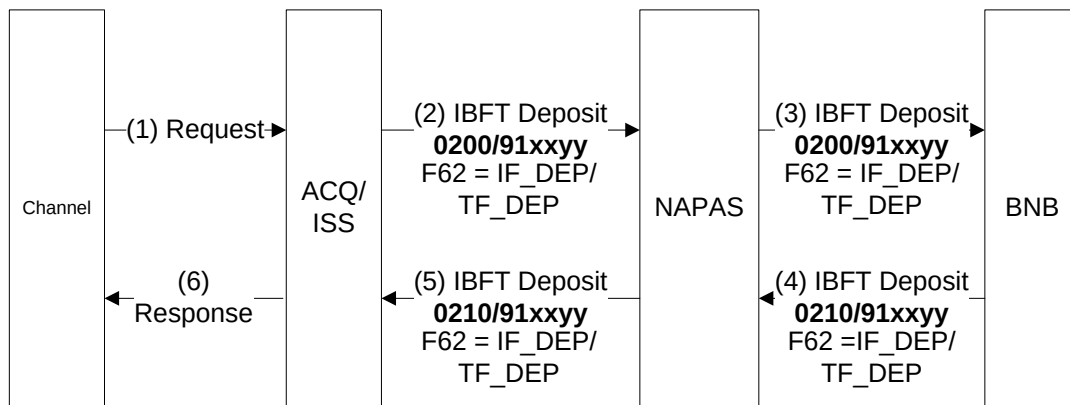


Figure 41. Transfer money to beneficiary card/account transaction process flow

Transaction flow detail:

Step	Transaction flow
1	The customer confirms fund transfer made on transaction Channel interface.
2	ACQ/ISS sends the fund transfer request to the beneficiary cardholder/ account (IBFT Deposit) for NAPAS.
3	NAPAS will forward the fund transfer request to the beneficiary cardholder/the account to BNB.
4	BNB receives the fund transfer request, shall perform the accounting for the beneficiary account and sends the notification of successful transfer

	transaction.
5	NAPAS sends the results of successful transfer transaction to ACQ/ISS.
6	ACQ/ISS shows the successful transfer notification to customers and finish the funds transfer to the beneficiary account flow.

4.3.2 Exceptions transaction flow

4.3.2.1 Regulation of automatic refund transaction process for customer

When ACQ/ISS processes the customer's refund request that does not get the response (time-out case) from NAPAS, ACQ/ISS should ensure to perform the following requirements:

- If the ACQ/ISS does not receive response from NAPAS about fund transfer results, ACQ/ISS does not refund the transaction.
- If the ACQ/ISS receives the successful response of NAPAS about fund transfer results, ACQ/ISS does not refund the transaction.
- If the ACQ/ISS receives the response of NAPAS about time-out fund transfer results (response code is 68), ACQ/ISS does not refund the transaction.
- If the ACQ/ISS receives an unsuccessful fund transfer response from NAPAS (the response code does not equal to 00 and 68, in which 00 is successful and 68 is time-out), ACQ/ISS will refund the transactions.

4.3.2.2 ACQ/ISS does not receive the results response of the beneficiary card holder/ account query request information from NAPAS

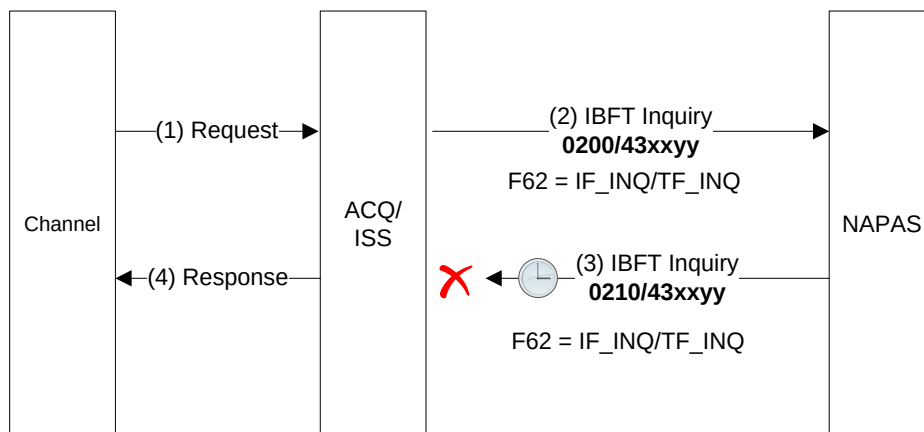


Figure 42. Transaction processing flow in case ACQ/ISS does not get the beneficiary card holder/ account information response from NAPAS

Transaction flow details:

Step	Transaction flow
1	Customers sends a query request information of beneficiary card holder/ account to ACQ/ISS.
2	ACQ/ISS sends the query information to the NAPAS.
3	NAPAS processes transaction and responds back to ACQ/ISS but ACQ/ISS have not received yet.
4	After a period of time-out regulations (this period is greater than a period of time-out regulations between NAPAS and BNB), ACQ/ISS sends notifications on the user interface to the customers that did not receive response.

4.3.2.3 NAPAS does not receive the beneficiary card holder/ account information query response from BNB

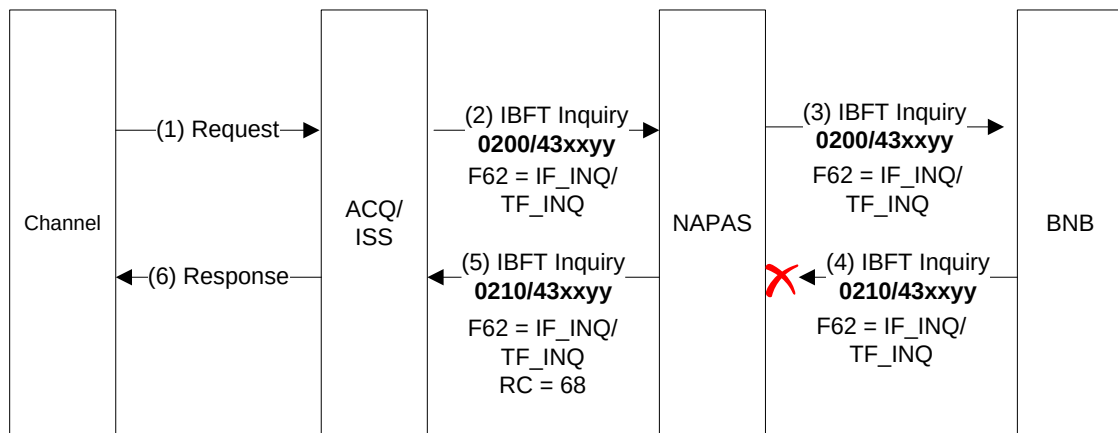


Figure 43. Transaction processing flow in case NAPAS does not get the beneficiary card holder/ account information response

Transaction flow detail:

Step	Transaction flow
1	Customers sends a beneficiary card holder/ account information query

	request to ACQ/ISS.
2	ACQ/ISS sends the query information to the NAPAS.
3	NAPAS sends the query information to BNB.
4	BNB responds back but NAPAS not received.
5	After time-out period, NAPAS systems will respond a 0210 transaction with RC = 68.
6	ACQ/ISS sends a notification to customers to report that not receive the query response.

4.3.2.4 ACQ/ISS does not receive the funds transfer response to the beneficiary card holder/ account from NAPAS.

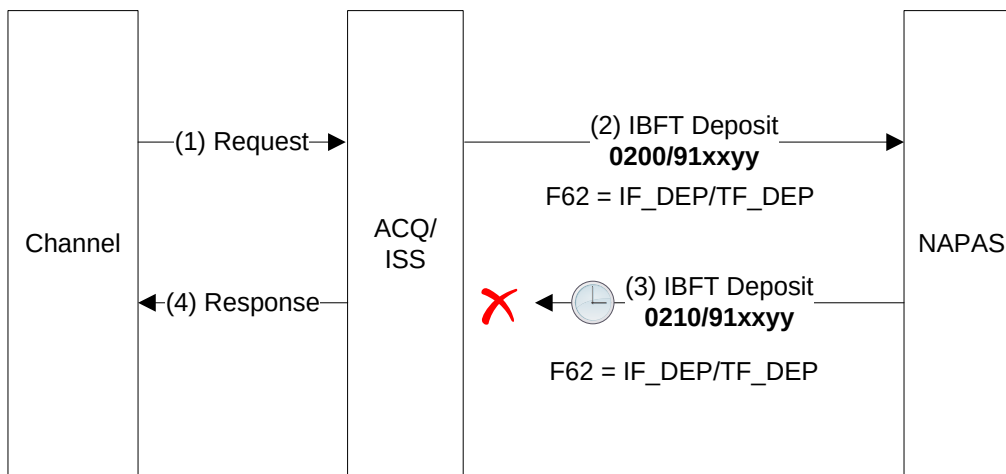


Figure 44. Transaction processing flow in case ACQ/ISS does not receive the fund transfer response to the beneficiary card holder / account from NAPAS

Transaction flow detail:

Step	Transaction flow
1	Customers sends a fund transfer request to the beneficiary cardholder/the account (IBFT Deposit) to ACQ/ISS.
2	ACQ/ISS sends the fund transfer information to NAPAS.

3	NAPAS processes transactions and responds to ACQ/ISS but ACQ/ISS does not receive.
4	After a period of time-out regulations, the ACQ/ISS sends a notification to inform the customer not receiving the fund transfer response from the NAPAS. For these transactions, the ACQ/ISS does not allow to refund to customer (see rules of automatic processing of refunds to customer above).

4.3.2.5 NAPAS does not receive the funds transfer response to the beneficiary card holder/ account from BNB.

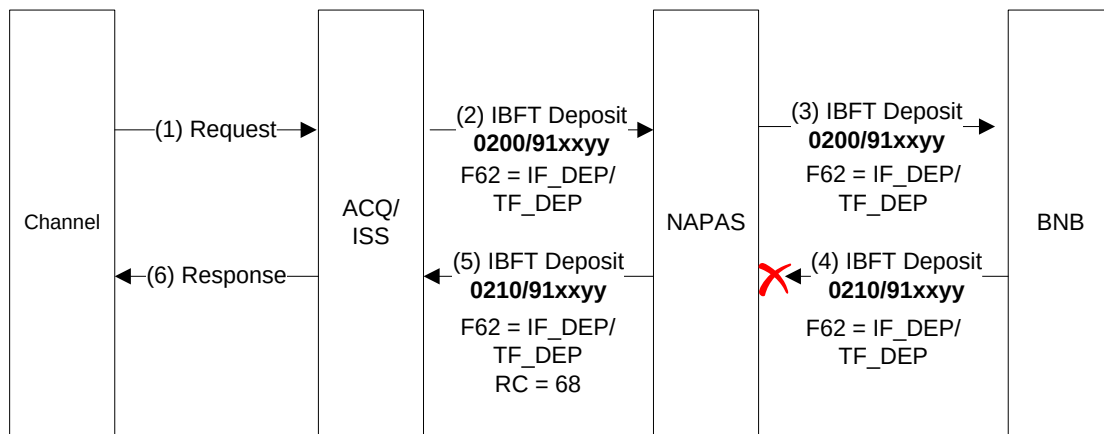


Figure 45. Transaction processing flow in case NAPAS does not receive the response of the funds transfer request to the beneficiary card holder/ account from BNB

Transaction flow detail:

Step	Transaction flow
1	Customers sends a fund transfer request to the beneficiary cardholder/ account to ACQ/ISS and ACQ/ISS forwards it to the NAPAS.
2	ACQ/ISS sends the fund transfer request to the beneficiary cardholder/ account to the NAPAS.
3	NAPAS sends the fund transfer request to the beneficiary cardholder/ account to the BNB.
4	BNB responds back but NAPAS does not receive.

5	After a period of time-out regulations, the funds transfer system of NAPAS response by a 0210 transaction with RC = 68.
6	ACQ/ISS sends a notification to inform customers not receiving the fund transfer response. For these transactions, the ACQ/ISS does not allow to refund to customer (refer to section "Regulation of automatic refund transaction process for customer")

4.4 Payment code service

4.4.1 Successful processing flow

Payment code service include two basic transaction flows:

- Create Payment code transaction flow :
 - Create payment code message.
- Cash withdrawal by Payment code flow:
 - Verify payment code message
 - Cash withdrawal by Payment code message

4.4.1.1 Payment code creation transaction flow

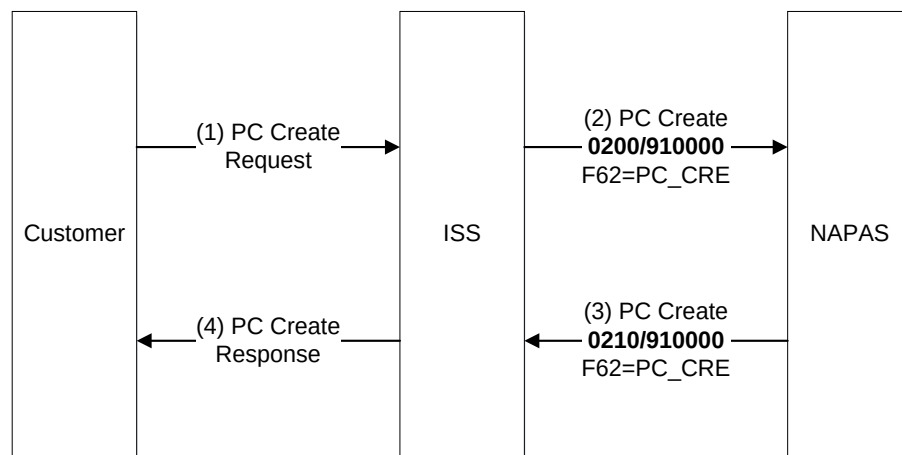


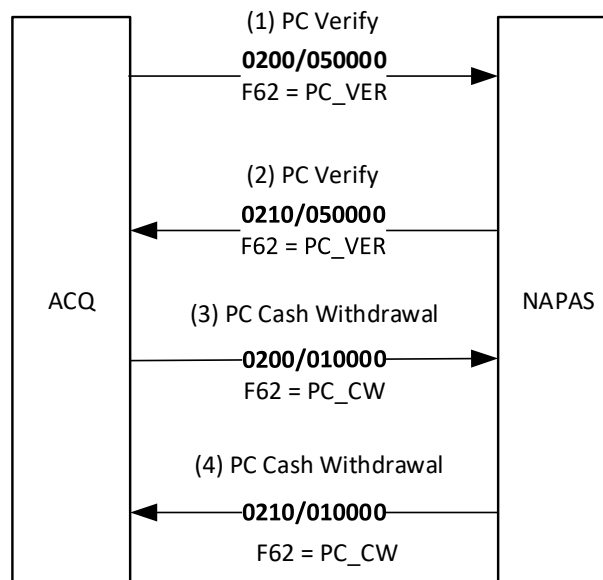
Figure 46. Payment code creation transaction flow

Transaction flow detail:

Step	Transaction flow
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1	The customer accesses to the payment channel of the ISS such as Internet Banking, Mobile Banking, ATM, Bank counters, etc for fund transfers, provide fund transfer information needed (the amount of the transaction, the phone number of beneficiary customer).
2	ISS receives a fund transfer request, checks the customer information and sends the Create Payment code request to the NAPAS.
3	NAPAS generates a PC and sends it back to the ISS
4	ISS forwards PC to customer (shown on ATM screen, Internet Banking, etc ...) and debit customer's account. Finish the source transfer flow, customer receives the PC and informs the code for the beneficiary customer.

4.4.1.2 Payment code cash withdrawal transaction flow



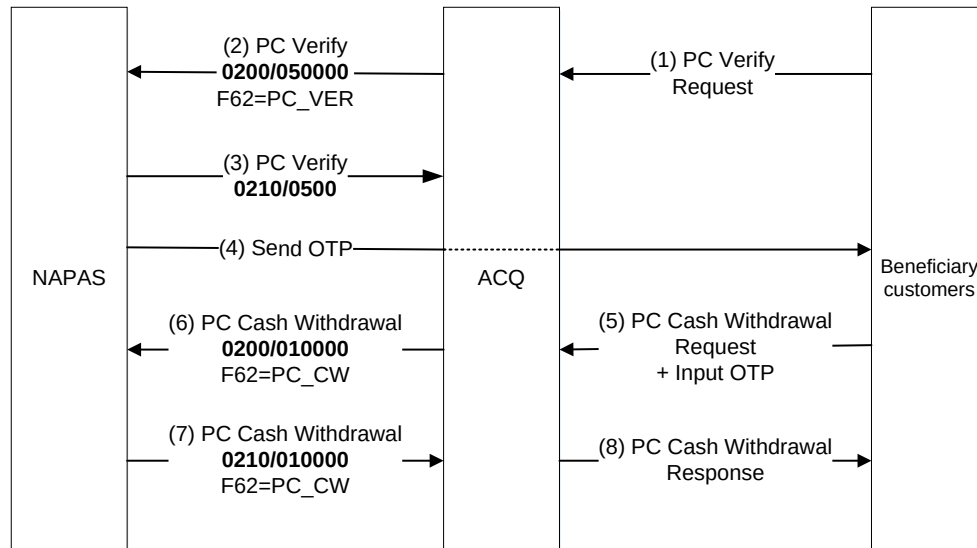


Figure 47. Payment code cash withdrawal transaction flow

Transaction flow detail:

Step	Transaction flow
1	Beneficiary customer comes to ATM of ACQ (Bank use Payment code service which NAPAS provide), enters the Payment code to cash withdrawal.
2	ACQ sends a Payment code (PC) verification message to NAPAS.
3	NAPAS verifies existing PC on NAPAS system and checks the status of PC. If the PC status is active and not locked, NAPAS will sends a successful Payment code verification response message, include the amount of Payment code for ACQ.
4	At the same time, NAPAS sends an OTP to the beneficiary customer via SMS.
5	Beneficiary customer enters an OTP on the ATM screen of ACQ to request a cash withdrawal transaction.
6	ACQ creates a payment code cash withdrawal message and sends it to NAPAS.
7	NAPAS verifies an OTP successfully, locks the PC and sends a successful response message to ACQ.

8	ACQ sends the command to the ATM to withdrawal money to customer.
---	---

4.4.2 Exception transaction flow

4.4.2.1 A Payment code creation request transaction is not successful (RC<>00)

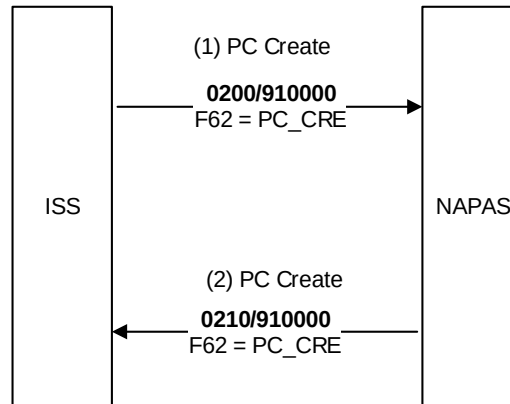


Figure 48. Unsuccessful Payment code creation transaction flow

Transaction flow detail:

Step	Transaction flow
1	ISS sends a Payment code creation message to NAPAS.
2	NAPAS processes unsuccessfully and sends the response message with error code to the ISS.

4.4.2.2 A Payment code verification transaction is not successful (RC<>00)

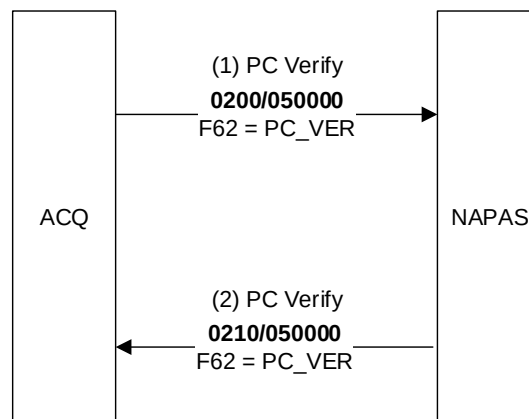


Figure 49. Unsuccessful Payment code verification transaction flow

Transaction flow detail:

Step	Transaction flow
1	ACQ sends a Payment code verification message to NAPAS.
2	NAPAS verifies a Payment code unsuccessfully and sends a response message with error code to ACQ.

4.4.2.3 A Payment code cash withdrawal transaction is not successful (RC<>00)

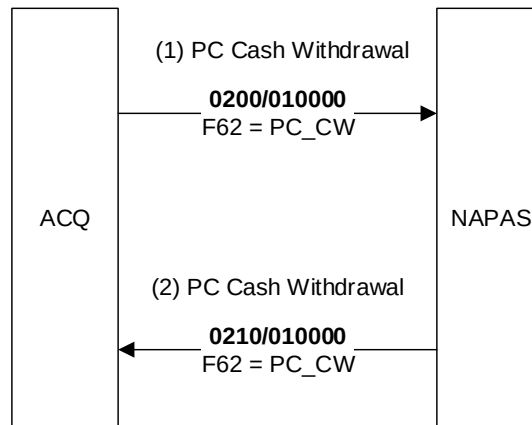


Figure 50. Unsuccessful Payment code cash withdrawal transaction flow

Transaction flow detail:

Step	Transaction flow
1	ACQ sends a Payment code cash withdrawal message to NAPAS.
2	NAPAS processes failed and send response message with error code to ACQ.

4.4.2.4 ISS does not receive a Payment code creation response (Time-out at the ISS)

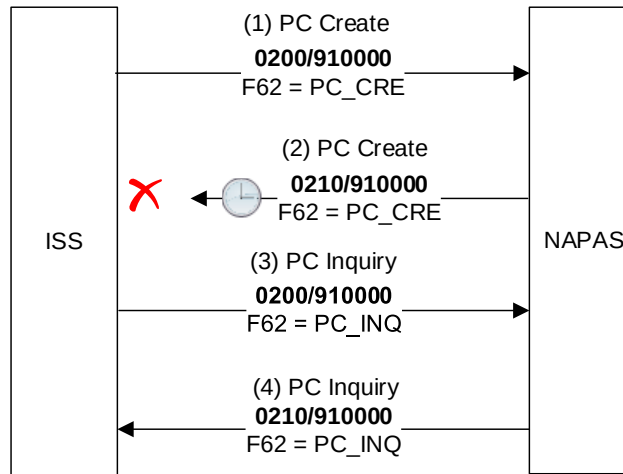


Figure 51. Transaction flow that is time-out at ISS

Transaction flow detail:

Step	Transaction flow
1	Sending customer uses Payment code service through the channel provided by the ISS. Customer will provide information such as phone number of beneficiary customer, the amount of the transaction. ISS sends a Payment code creation request with the customer information provided to NAPAS.
2	NAPAS sends a successful response message include Payment code to the ISS to send customer. ISS does not receive a response transaction from NAPAS.
3	After a period of time-out regulations, the ISS will send the Payment code query message to NAPAS.
4	NAPAS system will check: - If the Payment code that was created, NAPAS will send the successful response message with Payment code. - If the Payment code has not been created yet, NAPAS will send the unsuccessful response message to the ISS.

4.4.2.5 Unsuccessful cash withdrawal by Payment code failed in case of entering wrong OTP

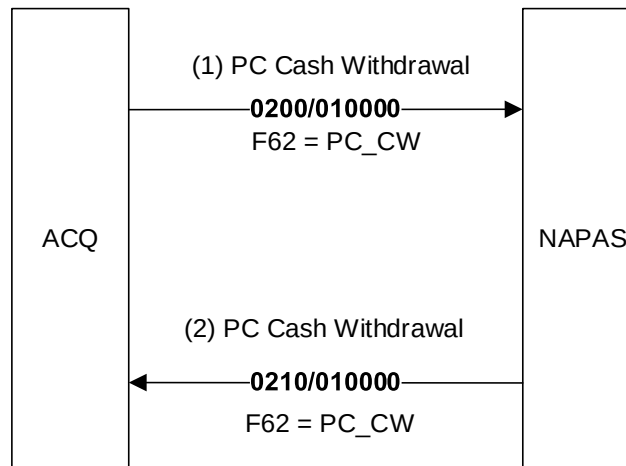


Figure 52. Wrong OTP processing flow

Transaction flow detail:

Step	Transaction flow
1	ACQ sends a Payment code cash withdrawal transaction including OTP to NAPAS.
2	NAPAS checks an OTP is not correct and sends a response message with error code as 55 to ACQ.

The system allows beneficiary customer to enter the wrong OTP maximum 03 times. In this case, beneficiary customer enters wrong OTP 03 times, NAPAS will lock the PC and sends a response message with RC = 75 to ACQ. Customers must make a dispute or wait after 03 days of PC expired, ISS will refund for sending customer.

4.4.2.6 ACQ does not receive Payment code cash withdrawal response message (Time-out at ACQ).

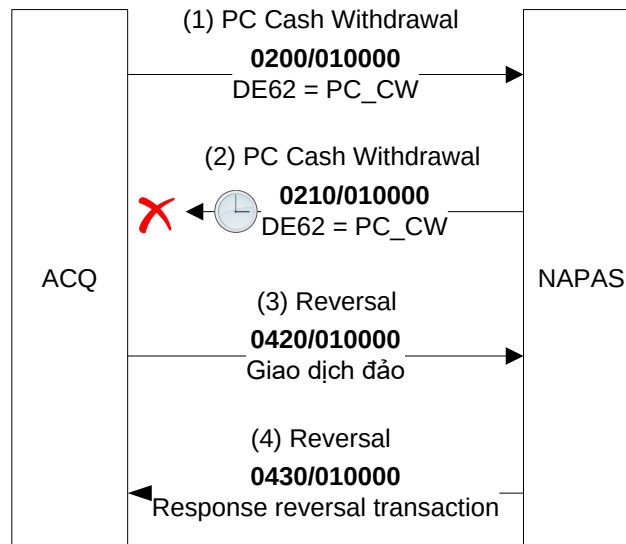


Figure 53. Transaction flow that is time-out at ACQ

Transaction flow detail:

Step	Transaction flow
1	Beneficiary customer enters an OTP code (an OTP is sent by NAPAS to beneficiary customer through SMS immediately after NAPAS verifies Payment code successfully) to request a cash withdrawal. ACQ sends a payment code cash withdrawal request message to NAPAS.
2	ACQ does not receive response from NAPAS.
3	After a period of time-out regulations. ACQ sends a reversal transaction to NAPAS.
4	NAPAS processes and sends a reversal response message to ACQ.

4.5 Tokenization service

4.5.1 Successful transaction flow

Tokenization service includes 02 transaction flows:

- General payment transaction flow
- Fast payment transaction flow

4.5.1.1 General payment transaction flow

General payment transaction flow includes 03 transaction:

- Verify cardholder information (Token Verify Card).
- Verify OTP (Token Verify OTP).
- Payment (Token Purchase).

General Tokenization transaction flow (including 03 types of processing transactions) and the card information is inputted in the Payment gateway's page are as follows:

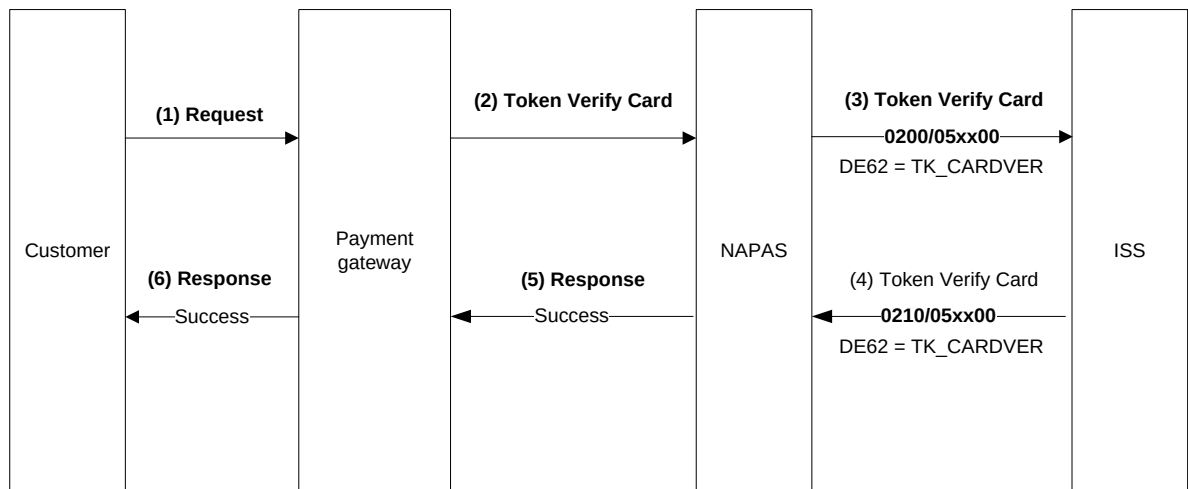


Figure 54. Cardholder information verification transaction flow (Token Verify Card)

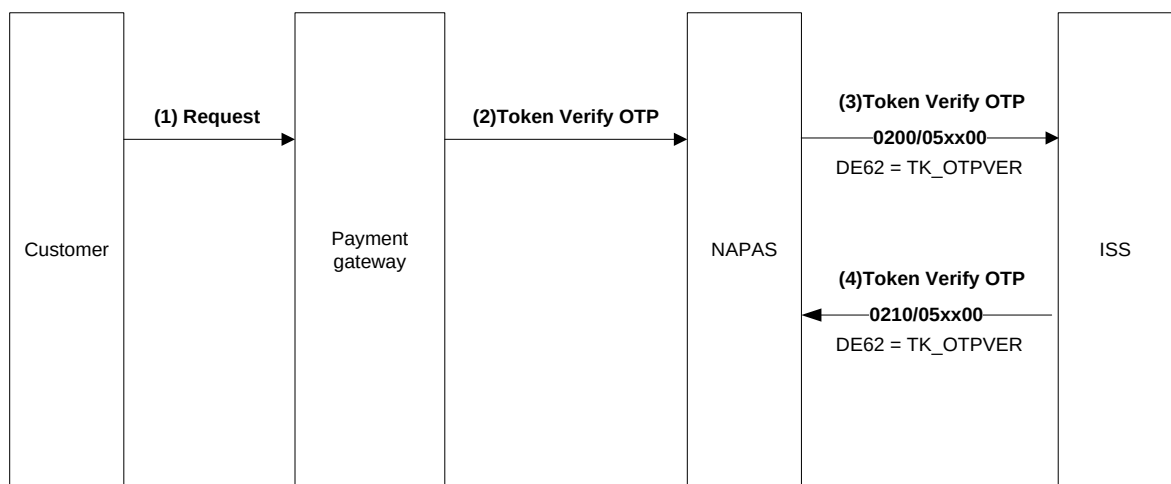


Figure 55. OTP verification transaction flow (Token Verify OTP)

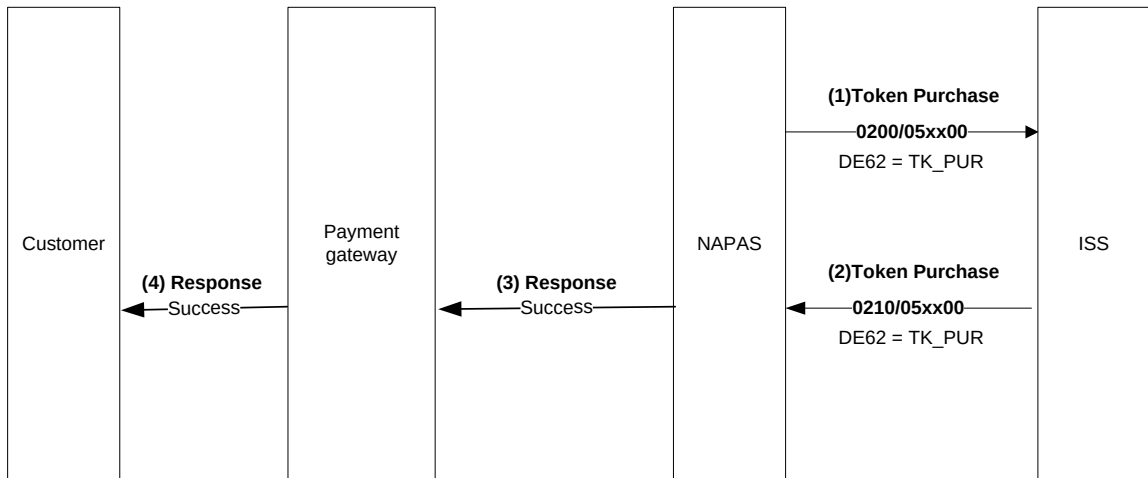


Figure 56. Payment transaction flow (Token Purchase)

Transaction flow detail:

Step	Transaction flow
Cardholder information verification transaction flow (Token verify card)	
1	Customer begins to pay for the purchase on web/interface of Payment Gateway. Customer enters cardholder information on Payment gateway interface.
2	Payment gateway sends a cardholder information verification request to NAPAS.
3	NAPAS receives and forwards to ISS to processing.
4	ISS verifies cardholder information successfully and ISS sends OTP to Customer. ISS responds the cardholder information verifying result to NAPAS.
5	NAPAS sends back the cardholder information response message to Payment gateway.
6	Payment gateway goes to OTP verification screen.
OTP verification transaction flow (Token Verify OTP)	
1	Customer enters an OTP code or CVV/CSC/PIN (if ISS supports) on payment interface.
2	Payment gateway sends an OTP verification request (Token Verify OTP) to NAPAS.

3	NAPAS receives OTP verification request (Token Verify OTP) and sends it to ISS to processing.
4	ISS verifies an OTP or CVV/CSC/PIN (if ISS authenticates) successfully and sends a result to NAPAS.
Payment transaction flow (Token Purchase)	
1	After receiving the results of successful OTP verification, NAPAS initiates a payment transactions (Token Purchase) and sends it to the ISS to processing the request.
2	ISS processes a requirement payment (debit customer account) and sends the successful payment result to NAPAS.
3	NAPAS sends a successful payment result to Payment gateway.
4	Payment gateways shows the successful result for customer and finish the online payment processing flow.

Note: in case of entering payment information on the payment gateway interface of NAPAS, the cardholder information and OTP information verification steps will display the notification on the interface NAPAS provides. The system only returns final payment results to Payment Gateway.

Tokenization transaction flow that performed a single Token Purchase transaction:

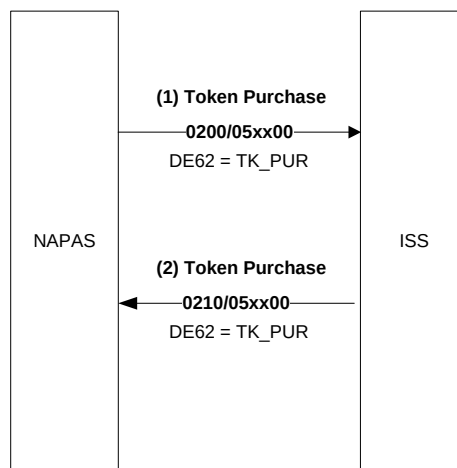


Figure 57. Payment transaction flow (Token Purchase)

Transaction flow detail:

Step	Transaction flow
Payment transaction flow (Token Purchase)	
1	NAPAs initiates a payment transaction (Token Purchase) and sends to the ISS to processing the request.
2	ISS processes a payment request (debit account of the customer's card) and sends the successful payment result to the NAPAS.

4.5.1.2 Fast payment transaction flow

Fast payment transaction flow have only one transaction:

- Fast payment transaction (Token Fast Purchase)

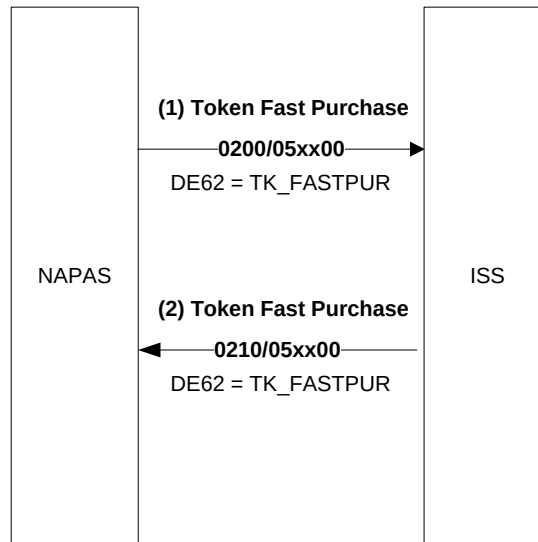


Figure 58. Fast payment transaction flow (Token Fast Purchase)

Transaction flow detail:

Step	Transaction flow
Fast Payment transaction flow (Token Fast Purchase)	
1	NAPAS initiates a payment transaction (Token Fast Purchase) include card holder information and sends it to the ISS to processing.
2	ISS checks and makes sure the transaction meets the necessary conditions, then verifies the card, account, authorizes transaction and

	sends a successful response to NAPAS.
--	---------------------------------------

4.5.2 Reversal transaction processing mechanism 0420/0430

0420/0430 Reversal transaction is used to reverse the payment transaction (Token Purchase) has done before. 0420/0430 Reversal transaction is initiated from the NAPAS.

When ISS receives the 0420 reversal message, the ISS will response immediately by 0430 message with RC = 00. A processing of the original transaction in the member bank system can be made later to improve the system performance.

Note: The NAPAS system does not process a reversal transactions has settlement date different from the current settlement date.

Reversal transaction processing flow as follows:

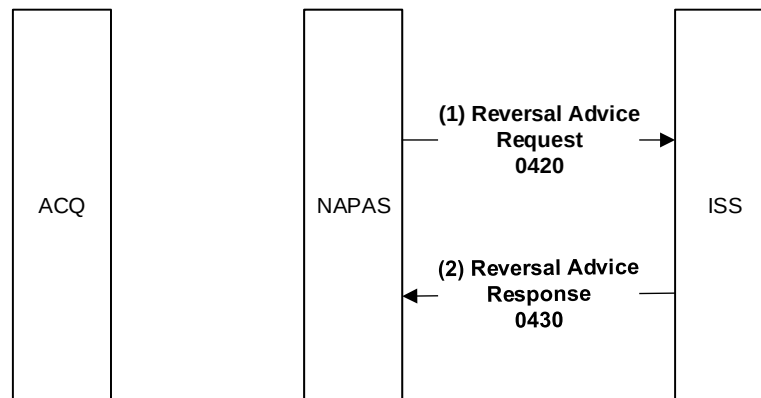


Figure 59. Reversal transaction flow

4.5.3 Exception transaction flow

4.5.3.1 Time-out cardholder information verification between NAPAS and ISS transaction

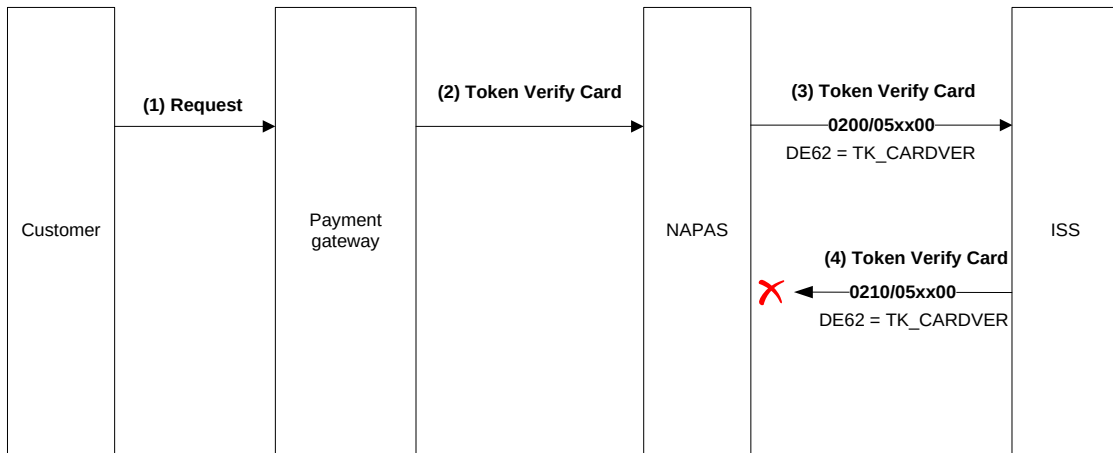


Figure 60. Time-out cardholder information verification between NAPAS and ISS transaction flow

Transaction flow detail:

Step	Transaction flow
1	Customer begins to pay for the purchase on web/interface of Payment Gateway. Customer enters a cardholder information (Token Verify Card) on Payment gateway interface.
2	Payment gateway sends a cardholder information verification message to NAPAS.
3	NAPAS receives and forwards it to ISS to process.
4	ISS responds a cardholder information verification results to NAPAS but NAPAS does not receive. The transaction is marked as time-out, finish transaction processing.

4.5.3.2 Time-out cardholder information verification between NAPAS and Payment gateway transaction

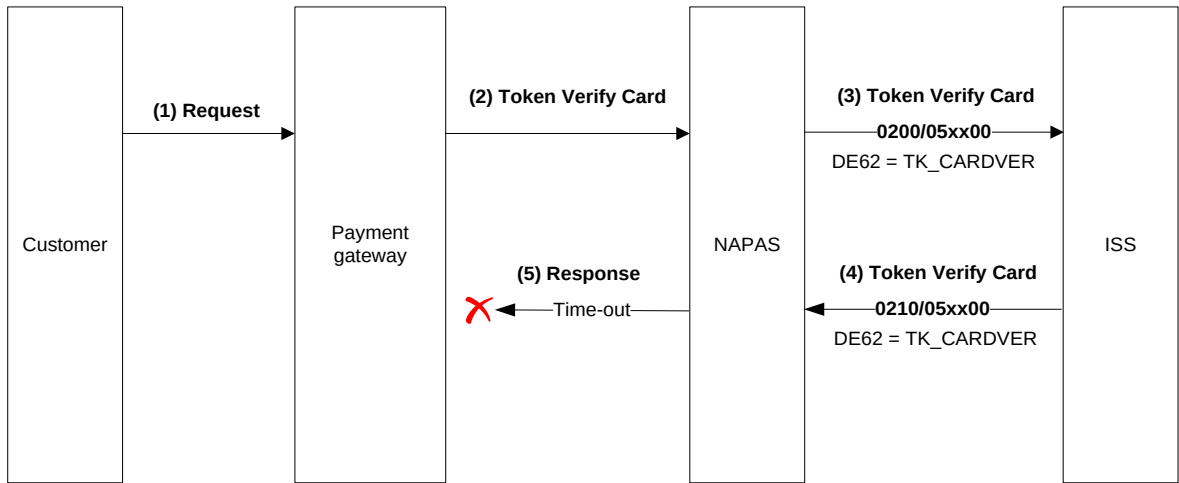


Figure 61. Time-out cardholder information verification between NAPAS and Payment gateway transaction flow.

Transaction flow detail:

Step	Transaction flow
1	Customer begins to pay for the purchase on web/interface of Payment Gateway. Customer enters a cardholder information (Token Verify Card) on Payment gateway interface.
2	Payment gateway sends a cardholder information verification message to NAPAS
3	NAPAS receives and forwards it to ISS to process.
4	ISS responds the cardholder information verification result to NAPAS.
5	NAPAS sends a cardholder information verification result for Payment gateway but for Payment gateway does not receive. Finish online transaction processing. Dispute transaction processing will follow with exception transaction process regulation for Payment Gateway.

4.5.3.3 Time-out OTP verification between NAPAS and ISS transaction.

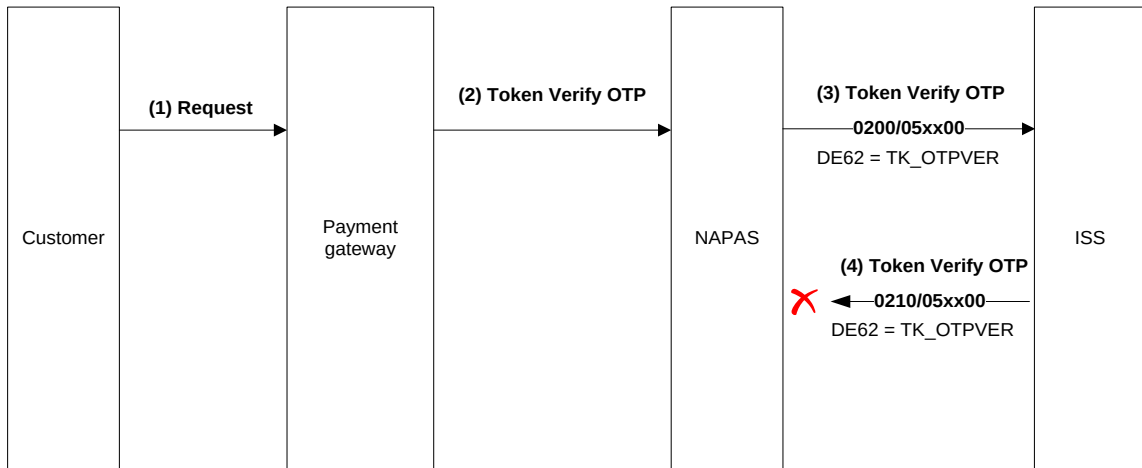


Figure 62. Time-out OTP verification between NAPAS and ISS transaction flow

Transaction flow detail:

Step	Transaction flow
1	Customer enters OTP code on the payment gateway interface.
2	Payment gateway sends an OTP verification request (Token Verify OTP) to NAPAS
3	NAPAS receives the OTP verification request (Verify OTP Token) and sends it to the ISS to perform processing the request.
4	ISS sends the response messages to NAPAS but NAPAS does not receive. The transaction is marked as time-out, finish transaction processing

4.5.3.4 Time-out verify OTP transaction between NAPAS and Payment gateway.

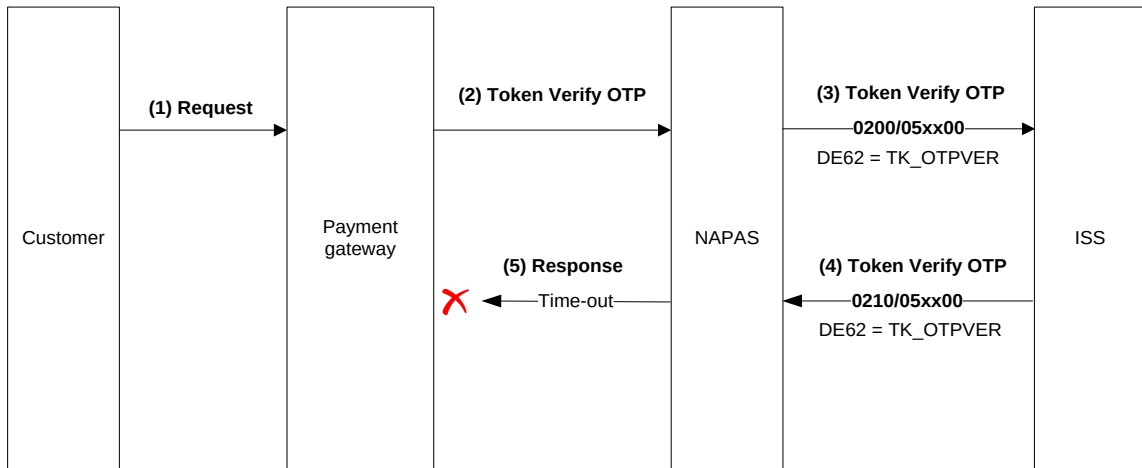


Figure 63. Time-out OTP verification between NAPAS and Payment gateway transaction flow

Transaction flow detail:

Step	Transaction flow
1	Customer enters an OTP code or CVV/CSC, PIN (if ISS supports) from ISS on the payment interface.
2	Payment Gateway sends an OTP verification request (Token Verify OTP) to NAPAS
3	NAPAS receives an OTP verification request (Verify OTP Token) and sends to the ISS to perform processing of the request.
4	ISS sends the successful OTP verification response messages to NAPAS.
5	NAPAS sends the successful response messages to Payment gateway but Payment gateway does not receive. Finish online transactions processing. Dispute transaction processing will follow with exception transaction process regulation for Payment Gateway.

4.5.3.5 Time-out purchase between NAPAS and ISS transaction.

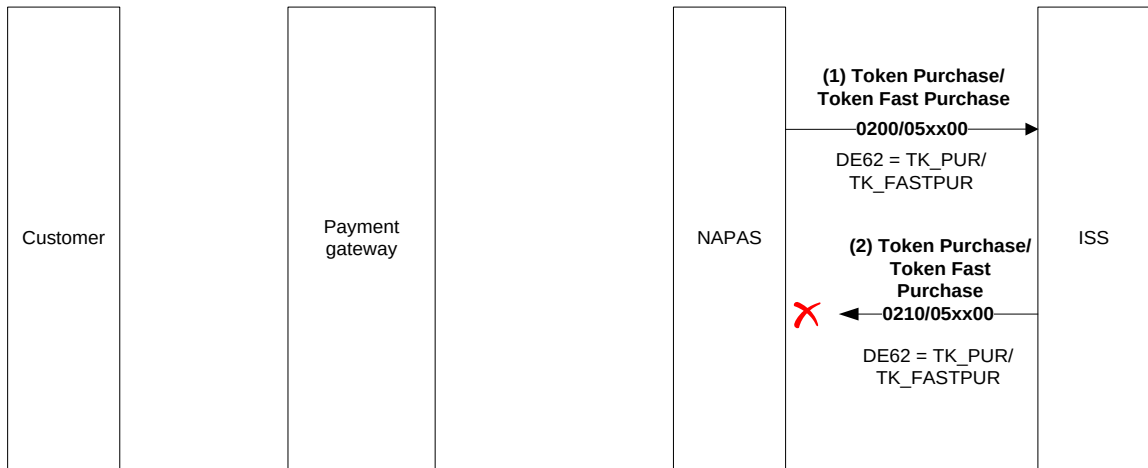


Figure 64. Time-out purchase between NAPAS and ISS transaction flow

Transaction flow detail:

Step	Transaction flow
1	NAPAS initiates a payment transactions (Token Purchase/Token Fast Purchase) and sends to the ISS to perform processing.
2	ISS sends the response messages to NAPAS but NAPAS does not receive. The transaction is marked as time-out, finish the purchase processing.

4.5.3.6 Time-out purchase between NAPAS and Payment gateway transaction

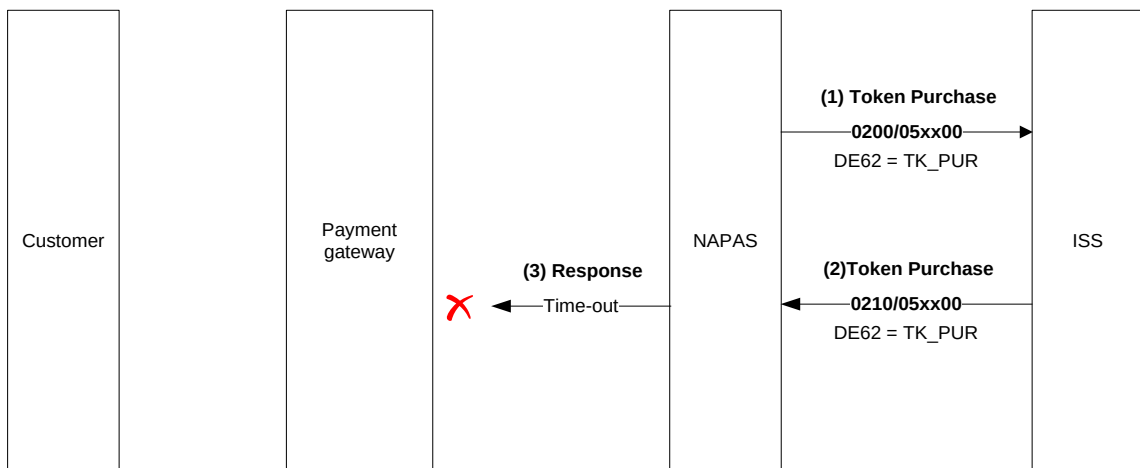


Figure 65. Time-out purchase between NAPAS and Payment gateway transaction flow

Transaction flow detail:

Step	Transaction flow
1	NAPAS initiates a payment transactions (Token Purchase) and sends it to the ISS to perform process.
2	ISS sends the response messages to NAPAS.
3	NAPAS sends the response messages to Payment gateway but Payment gateway does not receive. Finish online transactions processing. Dispute transaction processing will follow with exception transaction process regulation for Payment Gateway.

4.5.3.7 Expired OTP

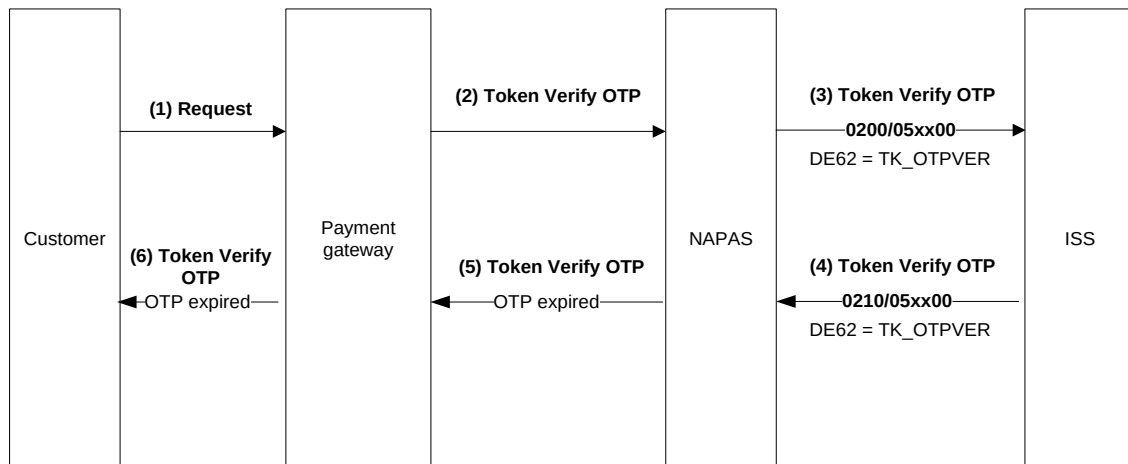


Figure 66. OTP expired processing flow

Transaction flow detail:

Step	Transaction flow
1	Customer enters an OTP code on the payment interface. System allows customer to enter OTP up to 03 times in a session.
2	Payment gateway sends an OTP verification request (Token Verify OTP) to NAPAS
3	NAPAS receives an OTP verification request (Verify OTP Token) and sends to the ISS to perform process.

4	ISS checks an OTP verification request but OTP is expired (time-out) and sends a response message to NAPAS with expired OTP status.
5	NAPAS sends the OTP verification response message with error response code to Payment gateway
6	Payment gateway sends the time-out response result to customer. ISS resends new OTP to customer and customer performs a transaction again.

4.5.3.8 Unsuccessful card information verification

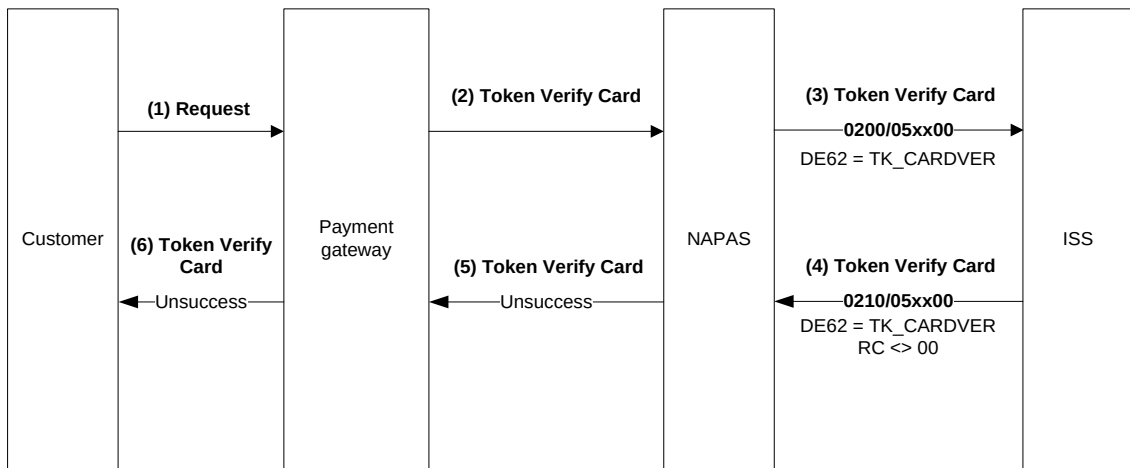


Figure 67. Unsuccessful card Information verification transaction flow

Transaction flow detail:

Step	Transaction flow
1	Customer enters cardholder information on the payment interface.
2	Payment Gateway sends a cardholder information verification request to NAPAS.
3	NAPAS forwards the receiving message to ISS.
4	ISS checks a cardholder information verification request and responds the unsuccessful result.
5	NAPAS sends a cardholder information verification response message to Payment gateway due to verify card information unsuccessfully from ISS.
6	Payment gateway sends back an unsuccessful response to customer.

4.5.3.9 Unsuccessful OTP verification

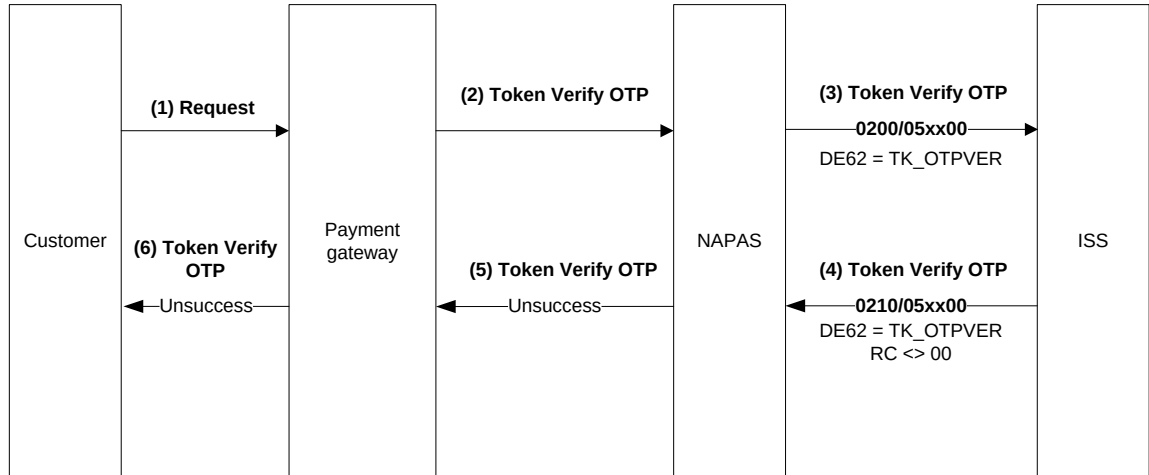


Figure 68. Unsuccessful OTP verification transaction flow

Transaction flow detail:

Step	Transaction flow
1	Customer enters OTP on Payment gateway interface. System allows customer to enter OTP up to 03 times in a session
2	Payment gateway sends an OTP verification request to NAPAS.
3	NAPAS forwards the receiving message to ISS.
4	ISS checks an OTP verification request and responds the unsuccessful OTP verification response.
5	NAPAS sends an unsuccessful OTP verification response to Payment gateway due to verify OTP unsuccessfully from ISS.
6	Payment gateway sends back unsuccessful response to customer.

4.5.3.10 Unsuccessful accounting

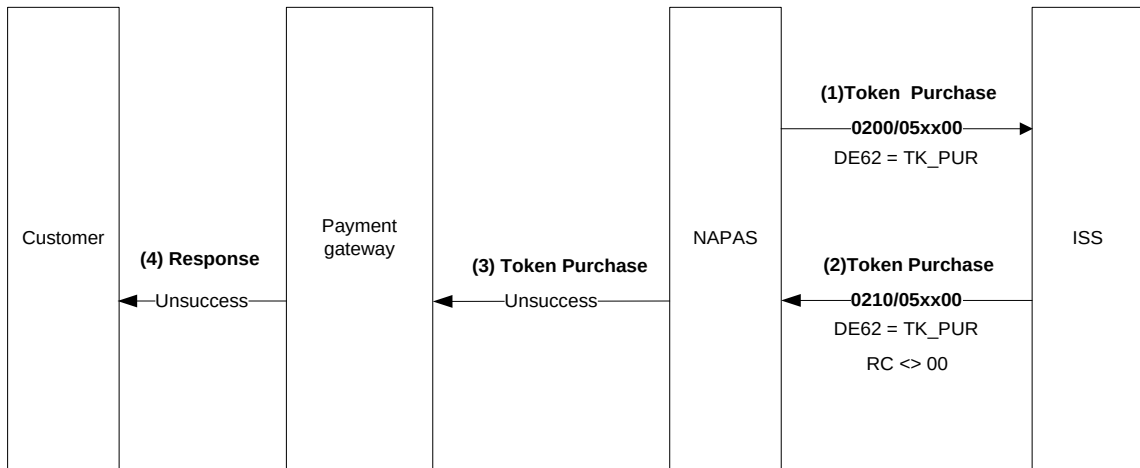


Figure 69. Unsuccessful accounting flow

Transaction flow detail:

Step	Transaction flow
1	After the customer sends the OTP verification request and NAPAS receives the successful OTP verification response from the ISS, NAPAS sends the purchase message (Token Purchase) to the ISS.
2	The ISS processes the request and returns unsuccessful accounting response.
3	NAPAS sends unsuccessful response to Payment Gateway.
4	Payment gateway sends the unsuccessful response to customers.

5 Cross-border card switching services processing flow

5.1 Cross-border card switching services on ATM/POS

Cross-border card switching services at ATM/POS allows cardholders of the Member Banks of NAPAS can perform transactions at ATM/POS in the Switching Organizations of other countries as well as at ATM/POS of foreign banks and the cardholder of the foreign bank of Switching Organizations of other countries can make transactions at ATM/POS of Member banks of NAPAS.

Cross-border card switching services at ATM/POS are deployed for the member banks on the basis of card switching services at ATM / POS between Member Banks and NAPAS. Including but not limited to the following transactions:

- Balance inquiry at ATM
- Cash withdrawal at ATM
- Balance inquiry at POS
- Purchase at POS

In which:

- With Switching Organization of China - China Union Pay (UPI), a service that allows member banks can accept both magnetic cards and chip cards of UPI at ATM / POS of member banks.
- With the other the Switching Organizations of other countries (as the Switching Organizations of APN such as MEPS (Malaysia), KTFC (Korea), NETS (Singapore), ITMX (Thailand); NCC|UC (Russia)), the Member Banks can deploy in both directions are acquiring banks and issuing banks on the ATM channel.
- The transaction and service model which connects to cross-border switching at the ATM / POS may be extended deployment on the platform of switching services ATM / POS between member banks and NAPAS.

Transaction processing flow of cross-borders card switching services at ATM / POS complies with the regulations on transaction processing flow of card switching services at ATM / POS reference described Section 4 - Annex VI.

5.2 Cross-Border Fund Transfer

Cross-Border Fund Transfer service (CBFT) allows banks's customers of the Switching Organizations of other countries to transfer money to the account/ card numbers of customers of NAPAS's member banks.

CBFT services for member banks deployed based on the service's interbank funds transfer (IBFT) between member banks and NAPAS, in which member banks acting as beneficiary bank.

Transaction processing flow of CBFT services follow the regulations on transaction processing flow of the IBFT service are described at Section 4 - Annex VI.

6 Network management transaction processing flow

6.1 Sign-on transaction (0800/001)

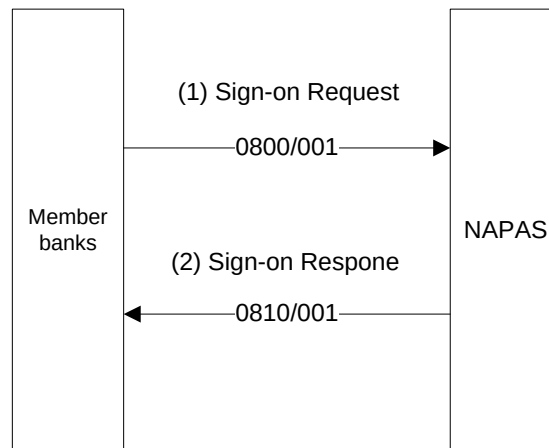


Figure 70. Sign-on transaction processing flow

Transaction flow detail:

Step	Transaction flow
1	Member Bank initiates a Sign-on message (0800/001) and sends it to NAPAS.
2	NAPAS sends a Sign-on response message (0810/001) and sends it to Member Bank.

BIN status of member bank will be set to "UP BIN" by NAPAS system when NAPAS receives a sign-on message from Member bank. NAPAS system just sends a financial message to Member bank if the BIN status of Member bank is "UP BIN".

6.2 Sign-off transaction (0800/002)

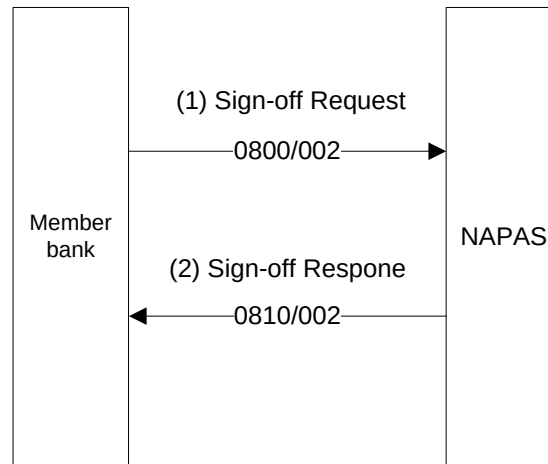


Figure 71. Sign-off transaction processing flow

Transaction flow detail:

Step	Transaction flow
1	Member Bank initiates a Sign-off message (0800/002) and sends it to NAPAS.
2	NAPAS send a Sign-off response message (0810/002) and sends it to Member Bank.

BIN status of member bank will be set to "DOWN BIN" by NAPAS system when NAPAS receives sign-off message from Member bank. If the BIN status of Member bank is "DOWN BIN" then there will be no financial transactions are sent to this Member bank. When Member bank run batch at the end of the day, or no need of receiving and processing transactions, Member bank can send Sign-off message to NAPAS system to suspend service.

6.3 Key exchange transaction (0800/161)

6.3.1 Key exchange request of Member bank

Member Bank may ask NAPAS to perform key exchange transactions whenever necessary. Key exchange transactions should be done regularly every day at 01:00 AM.

Member Bank uses only one key pair and MAK (key pairs are made during the most recent key exchange) or 01 ZPK or 01 MAK (depending on deployment services for NAPAS) to perform assigned translation for NAPAS.

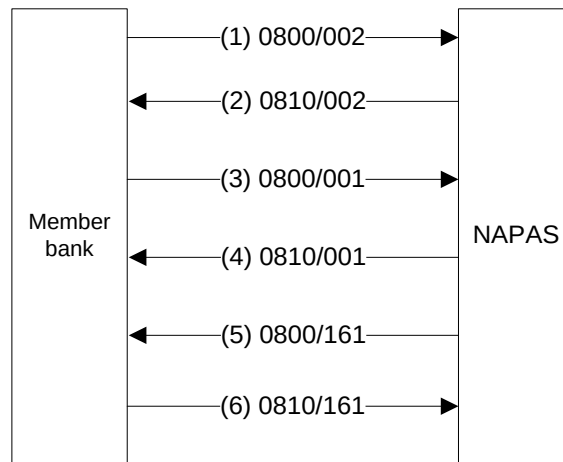


Figure 72. Key exchange request of Member bank flow

Transaction flow detail:

Step	Transaction flow
1	Member Bank initiates a Sign-off message (0800/002) and sends it to NAPAS.
2	NAPAS responds a Sign-off response message (0810/002) and sends it to Member Bank.
3	Member Bank initiates a Sign-on message (0800/001) and sends it to NAPAS.
4	NAPAS responds a Sign-on response message (0810/001) and sends it to Member Bank.
5	NAPAS initiates a Key exchange message (0800/161) and sends it to Member Bank.
6	Member Bank responds a Key exchange response message (0810/161) to NAPAS confirm using new key.

After processing Sign-off messages and a Sign-on messages is received from Member bank successfully, NAPAS automatically sends a Key exchange messages (0800/161) containing new keys to Member bank to perform key exchange.

In case the member bank uses both key ZPK and MAK, the NAPAS system will automatically send two Key exchange messages (0800/161) containing new key for member bank to implement both key exchange (a message include ZPK, a message include MAK).

6.3.2 Exchange key message from NAPAS

NAPAS can proactively exchange key by sending the Key exchange message (0800/161) to the Member bank.

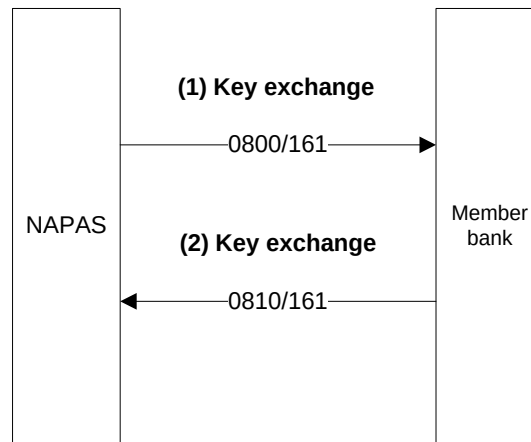


Figure 73. Exchange key message flow from NAPAS

Transaction flow details:

Step	Transaction flow
1	NAPAS initiates a Key exchange message (0800/161) and sends it to Member Bank.
2	Member Bank responds a response message (0810/161) to NAPAS for confirm using new key.

NAPAS automatically sends a Key exchange messages (0800/161) that contains new key for Member bank to exchanging key.

In cases Member bank uses both MAK and ZPK, NAPAs system can automatically send two Key exchange (0800/161) messages containing new key to Member bank to performing the exchange of the two keys (the message include ZPK, the message include MAK) or sends a key exchange message (0800/161) that contains new keys for Member bank to perform a key exchange.

6.4 Echo-test transaction (0800/301)

Echo-Test (0800/301) message is used to check the connection between the Member bank and NAPAS system. Every 05 minutes NAPAS and member bank send echo-test messages per time.

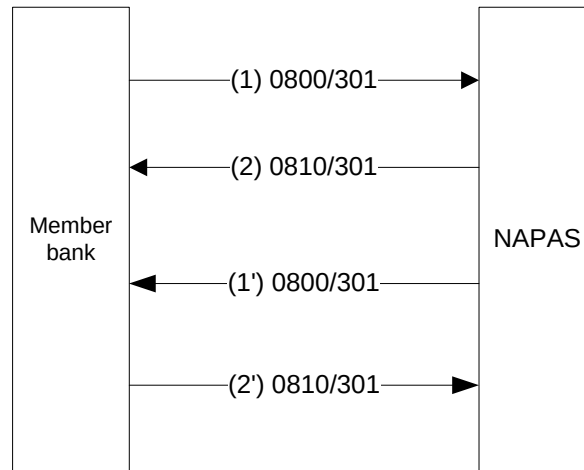


Figure 74. Echo-test transaction flow

Transaction flow detail:

Step	Transaction flow
1	Member bank initiates an Echo-test message (0800/301) and sends it to NAPAS.
2	NAPAS responds an Echo-test response message (0810/301) to Member bank.
1'	NAPAS initiates an Echo-test message (0800/301) and send it to Member bank.
2'	Member bank responds an Echo-test response message (0810/301) to NAPAS.

7 Regulation on mandatory implementation of transaction types

Transaction type	Mandatory implementation (Yes/No)
Domestic card switching service	
Switching service at ATM	
Cash Withdrawal	Yes
Balance inquiry	Yes
Mini-statement	Yes
InTra-bank Fund Transfer	No
PIN Change	No
Switching service at POS	
Balance inquiry	No
Purchase	Yes
Void	Yes
PIN change	No
Online switching service (Ecom)	
Online payment (Ecom)	No
Inter-Bank Fund Transfer 24/7 service (IBFT)	
Inter-Bank Fund Transfer	No
Payment code service	
Payment code	No
Tokenization service	
Tokenization	No
Cross-border card switching service	
Cross-border card switching at the ATM/POS	No
Cross-border Fund Transfer	No
Network management transaction type	
Sign-on/ Sign-off	Yes

Key exchange	Yes
Echo-test	Yes

8 Regulation on Time-out period

Time-out period is the maximum period since a request message (Request) which is sent to until the time of receiving a response message (Response).

After Time-out period:

- If NAPAS does not receive the response message from ISS, NAPAS will perform exception about time-out period corresponding to each transaction type of each service (example, with cash withdrawal at the ATM transaction, NAPAS will initiate a response message with time-out code to ACQ, initiates simultaneously a reversal message to ISS).
- If ACQ does not receive a response message from NAPAS, ACQ will perform exception about time-out period corresponding to each transaction type of each service (example, with cash withdrawal at the ATM transaction, ACQ will initiate a reversal message to NAPAS).
- If transaction channel does not receive any response from ACQ, these channels will process exceptions regarding time-out cases corresponding to transaction types of each service (example, with cash withdrawal at ATM, ATM will initiate a reversal request and send it to ACQ).
- If ISS sends the response message after time-out period, NAPAS's system will not process the message and consider it as a late response message (Late response).

Time-out period between the parties involved in the model of card switching service connection of NAPAS are set as follows:

Time-out Period Regulation		
No.	Request – Response	Time-out period
1	NAPAS – ISS/BNB	15s
2	ACQ – NAPAS	20s
3	Transaction Channel - ACQ	>20s

In that:

- Time-out period between NAPAS and ISS/BNB (or institution receives a response message) is set default as 15 seconds and could be customized to meet the actual needs.
- Time-out period between ACQ (or institution sends a request message) and NAPAS is set default as 20 seconds and could be customized to meet the actual needs.
- Time-out period between transaction channel and ACQ (or institution sends a request message) is set default as greater than 20 seconds and could be customized to meet the actual needs.

Transaction flow below describes the settings and time-out processing:

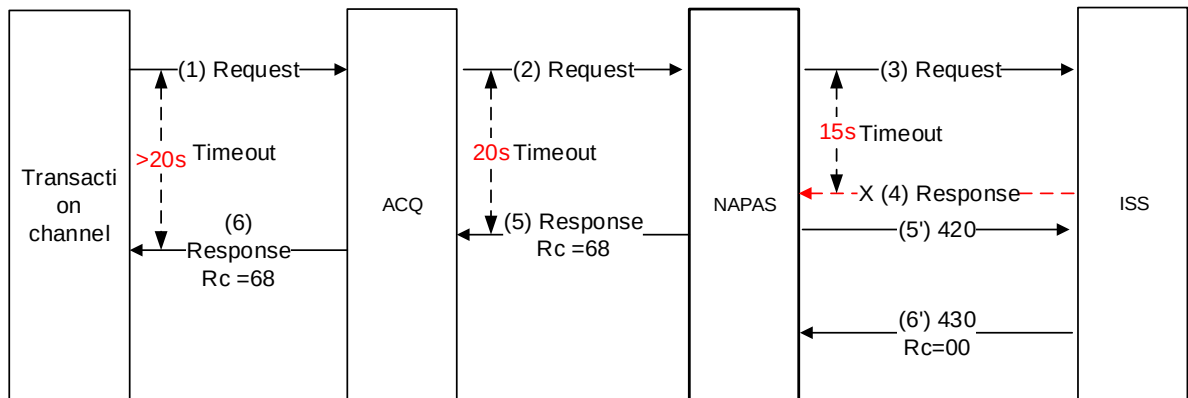


Figure 75. Time-out transaction processing flow

9 Regulation on DE #15 field (Settlement date) processing

With NAPAS card switching system, transactions are processed all days in a week (24x7), include day-off and holidays.

NAPAS will determine settlement day value based on cut-off time of system regulation (DE#15 value change each day) and fills into DE#15 (Settlement date) for all transactions which send to and send from NAPAS. Any value is filled from member bank, it will be replaced by settlement date which NAPAS specified.

- NAPAS will fill into DE#15 field in the message (0200) before forwarding to ISS.

- When NAPAS receives the response message (0210), NAPAS will use the same settlement date which filled in the message (0200) to send to ISS, fill into DE#15 field in the response message (0210) and sends it to ACQ

NAPAS uses DE # 15 to exporting the reconciliation files for the Member bank.

10 Effective document

The NAPAS Technical Specification for switching service takes effect from 01st January, 2017.

11 Document management

Internal reference document:

No.	Document Name
1	BANKNETVN Technical Specification Document (version 1.6.1).
2	SMARTLINK Technical Specification Document (version 2.0).
3	Regulation for Member Organization Participating on the SMARTLINK System.
4	Regulation for Member Organization Participating on the BANKNETVN System.
5	Regulation for Business Settlement and Reconciliation Process on the BANKNETVN System (released at 28th February, 2013).

External reference document:

No.	Document Name
1	ISO 8583-1987 Technical Specification Document.
2	Technical Specification Documents of VISA Organization.
3	Technical Specification Documents of MASTERCARD Organization.
4	Technical Specification Documents of UPI Organization.